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(54) **IMAGING LENS ASSEMBLY, IMAGING
DEVICE AND MOBILE TERMINAL**

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G02B 13/00 (2006.01)
G02B 9/62 (2006.01)

(52) **U.S. Cl.**
CPC **G02B 13/0045** (2013.01); **G02B 9/62**
(2013.01)

(58) **Field of Classification Search**

CPC G02B 13/0045; G02B 9/62
USPC 359/713
See application file for complete search history.

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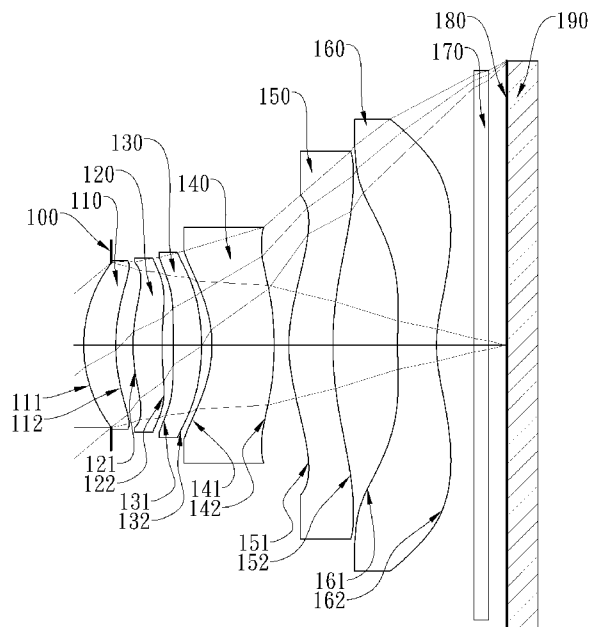
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(57) **ABSTRACT**

An imaging lens assembly includes, in order from an object side to an image side, a first lens element, a second lens element, a third lens element, a fourth lens element, a fifth lens element, and a sixth lens element. The first lens element has refractive power. The second lens element has refractive power. The third lens element has positive refractive power. The fourth lens element has negative refractive power. The fifth lens element with refractive power has an image-side surface being concave in a paraxial region thereof. The sixth lens element with refractive power has an image-side surface being concave in a paraxial region thereof, wherein the image-side surface of the sixth lens element has at least one convex shape in an off-axis region thereof. The imaging lens assembly has a total of six lens elements with refractive power.

28 Claims, 20 Drawing Sheets



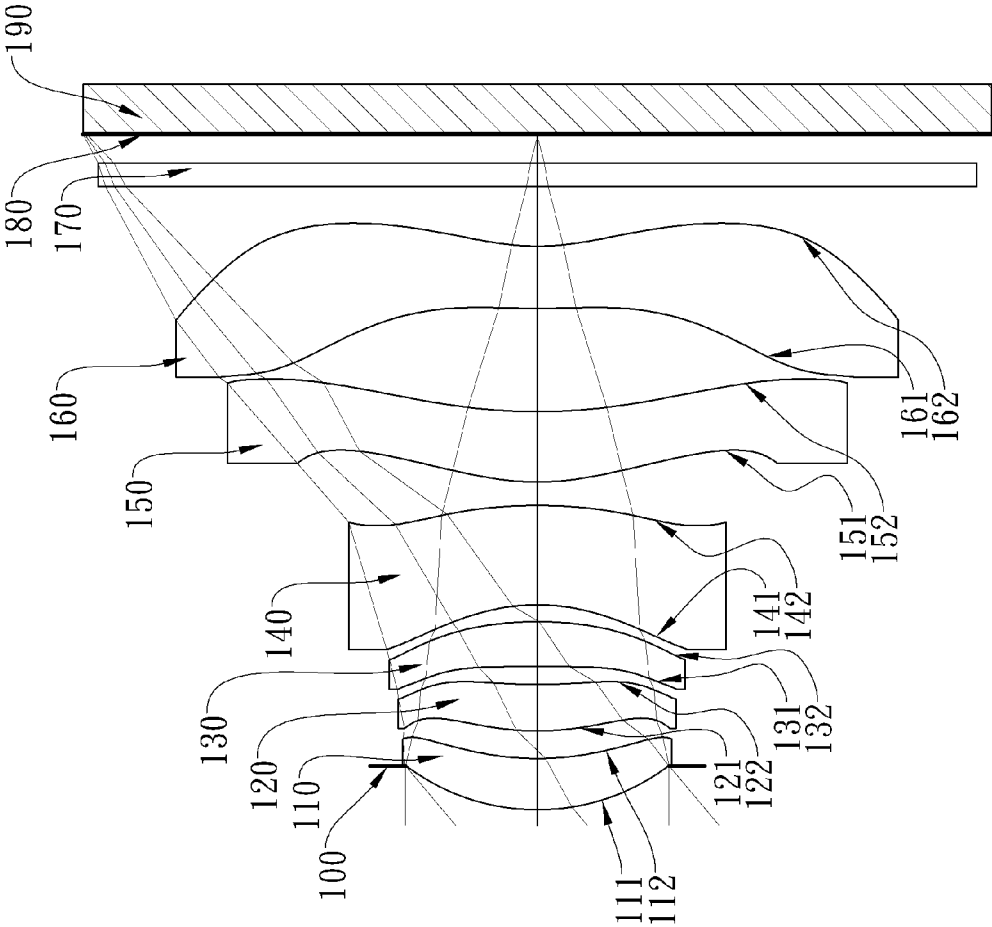


Fig. 1A

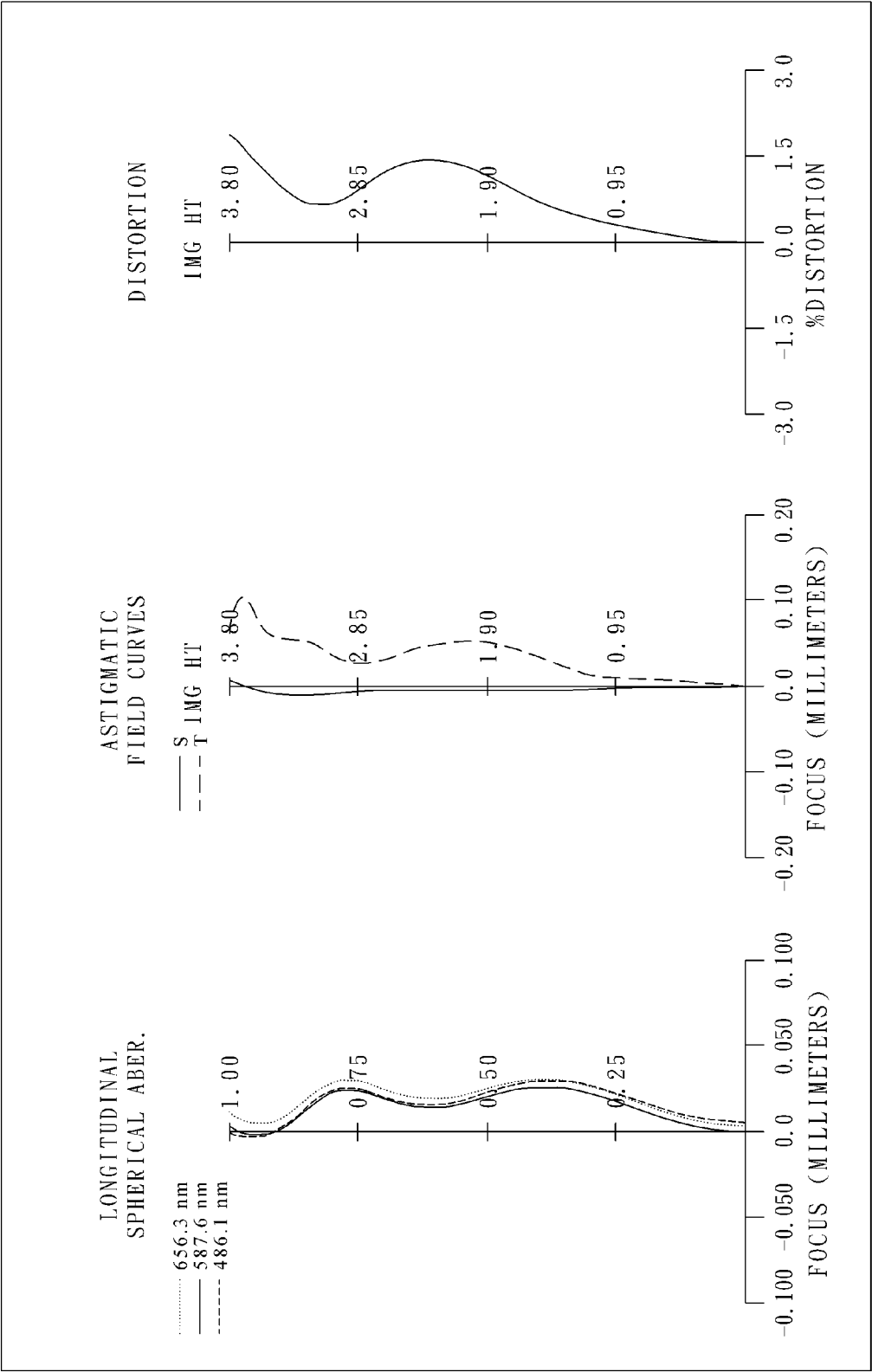


Fig. 1B

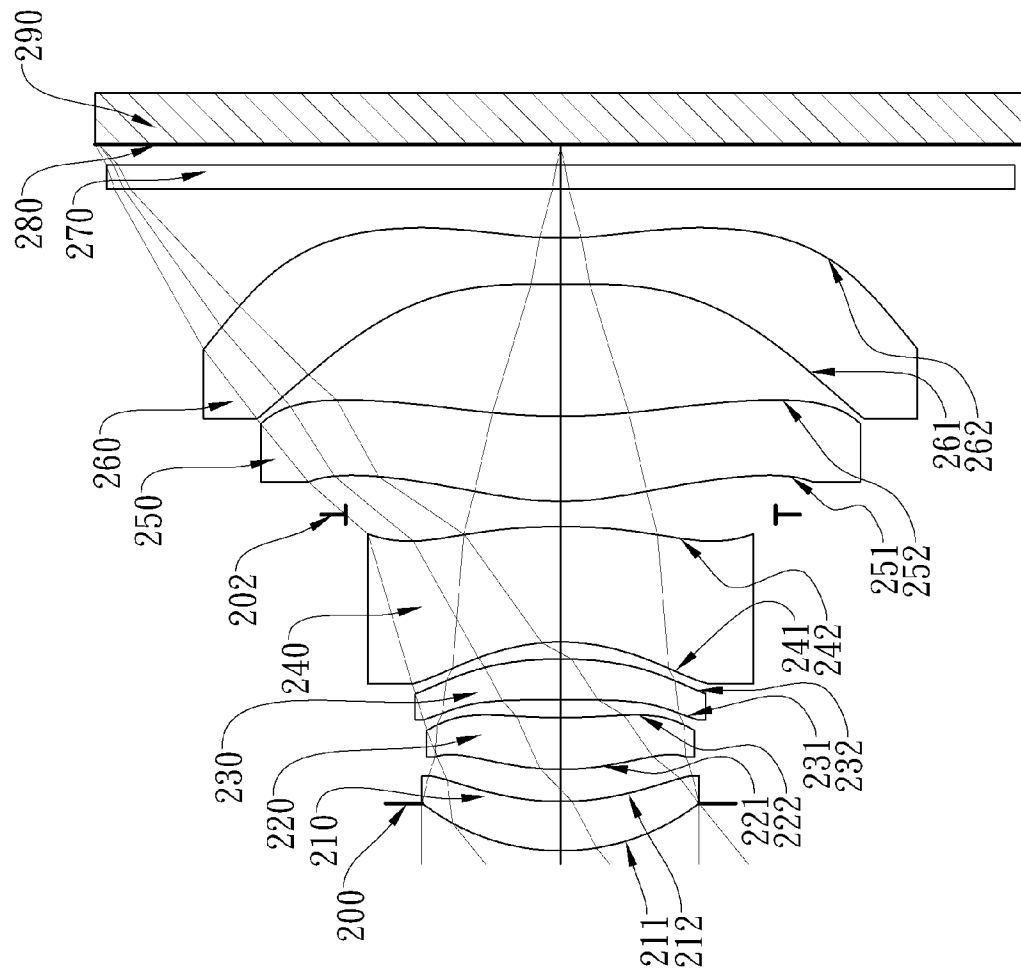


Fig. 2A

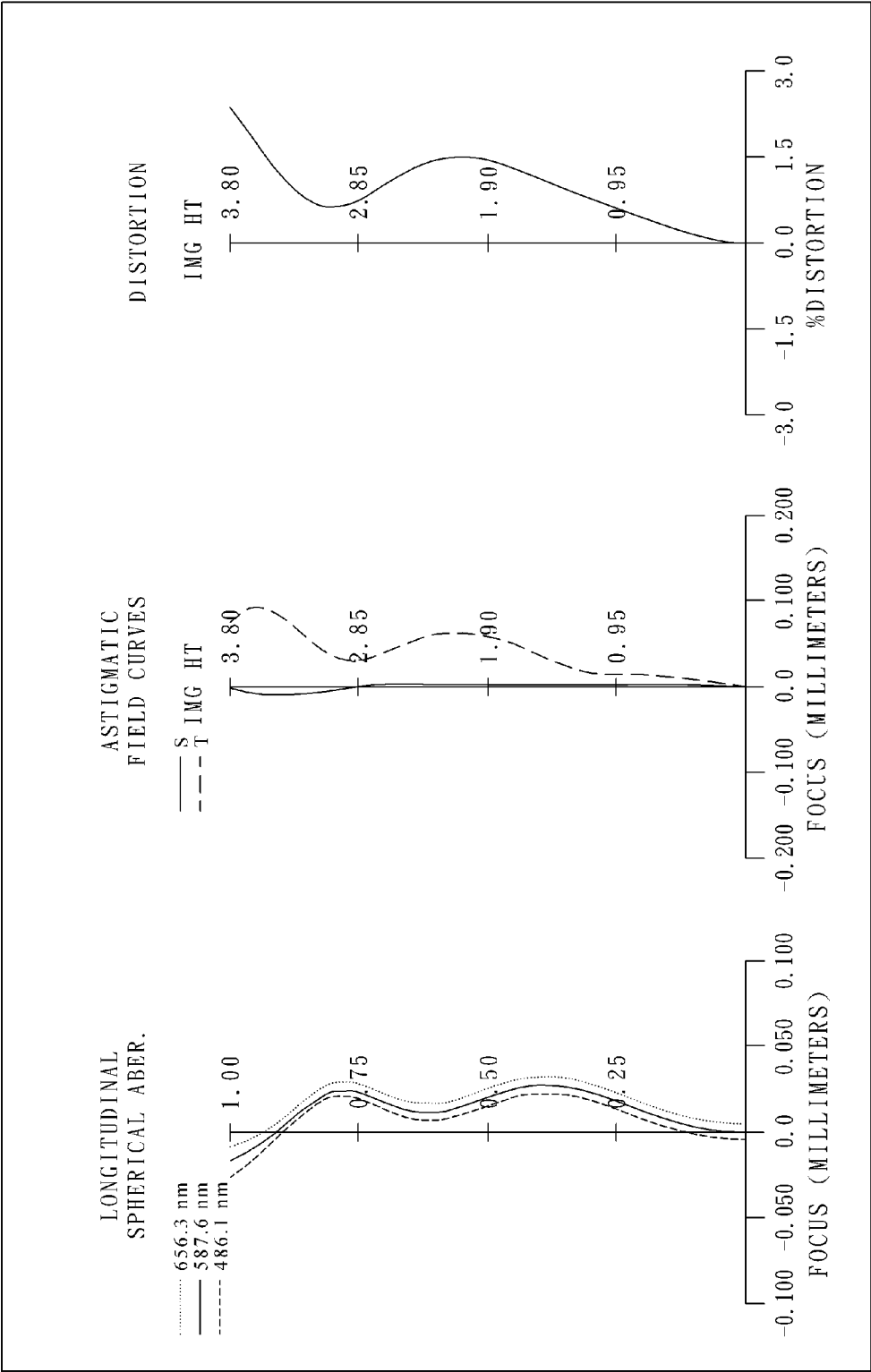


Fig. 2B

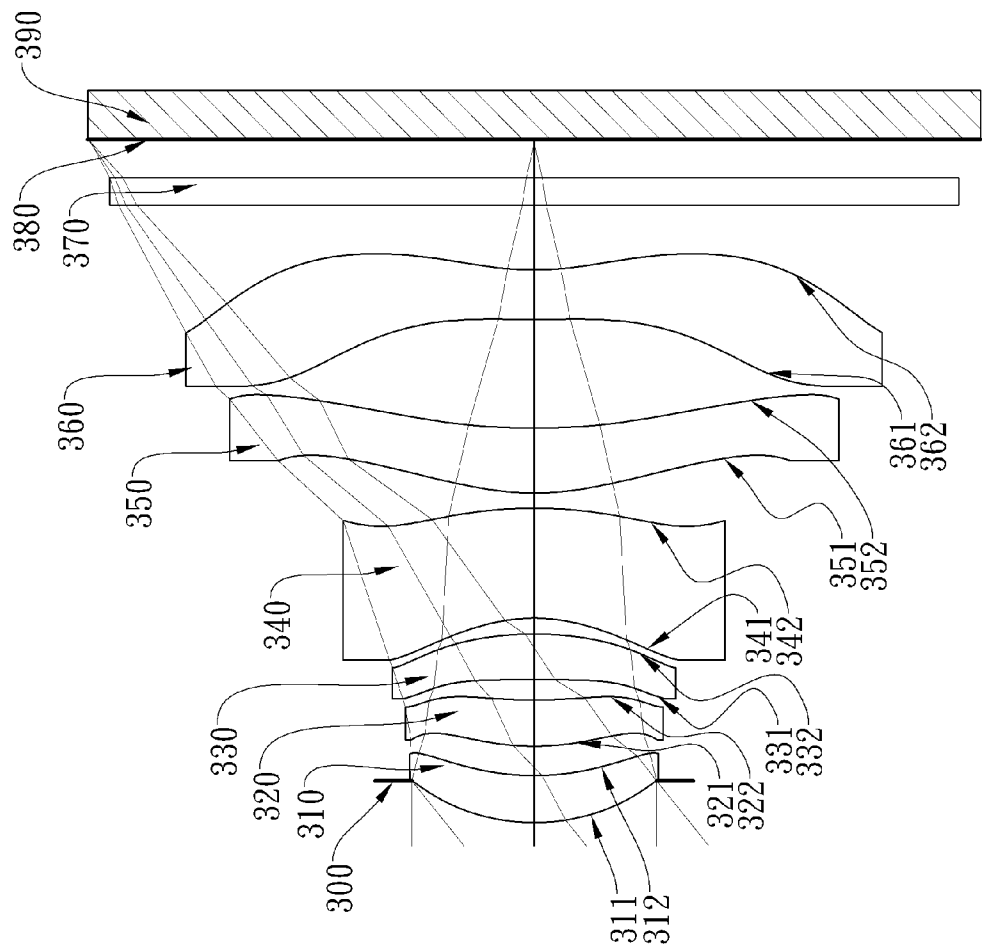


Fig. 3A

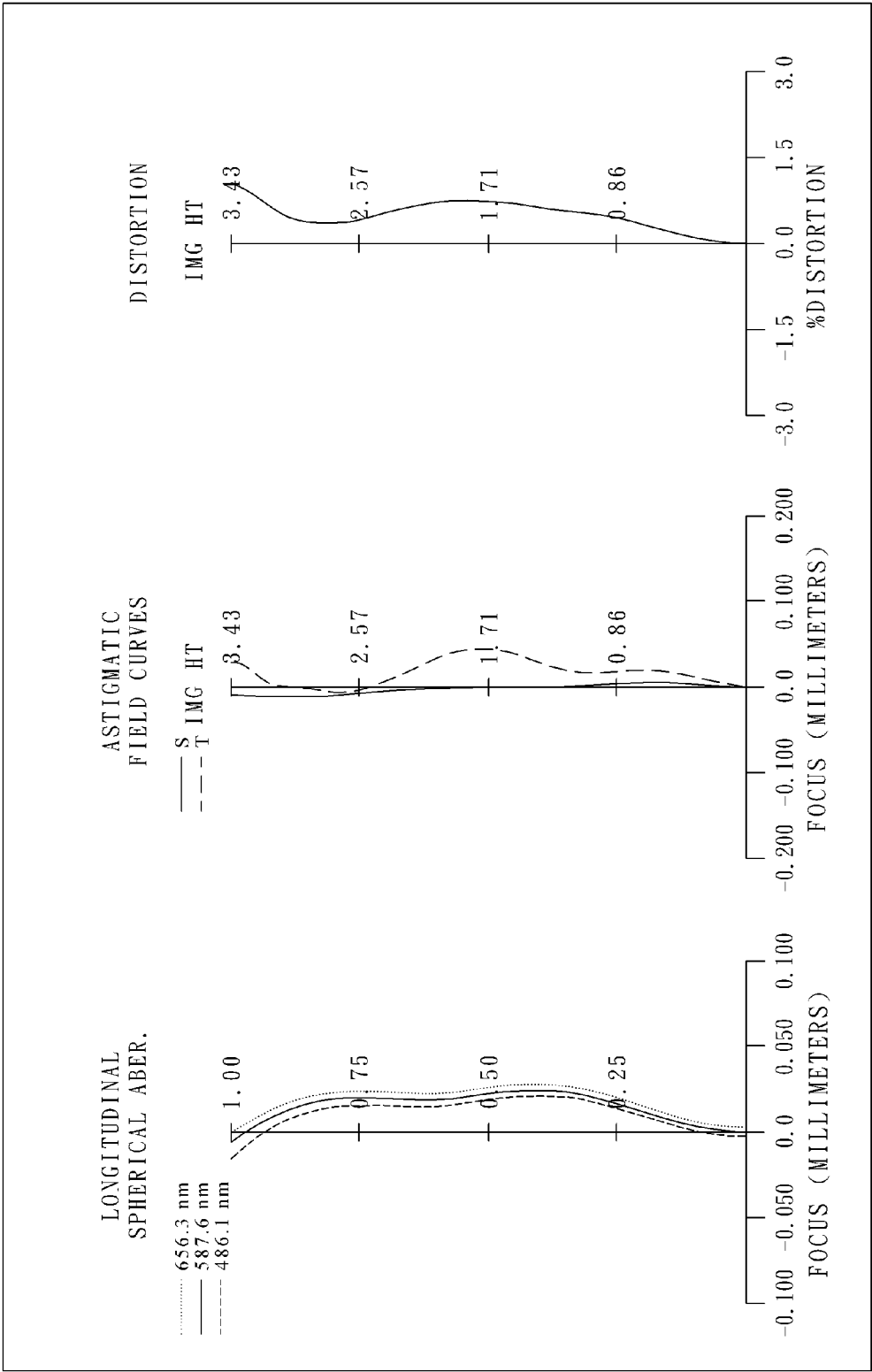


Fig. 3B

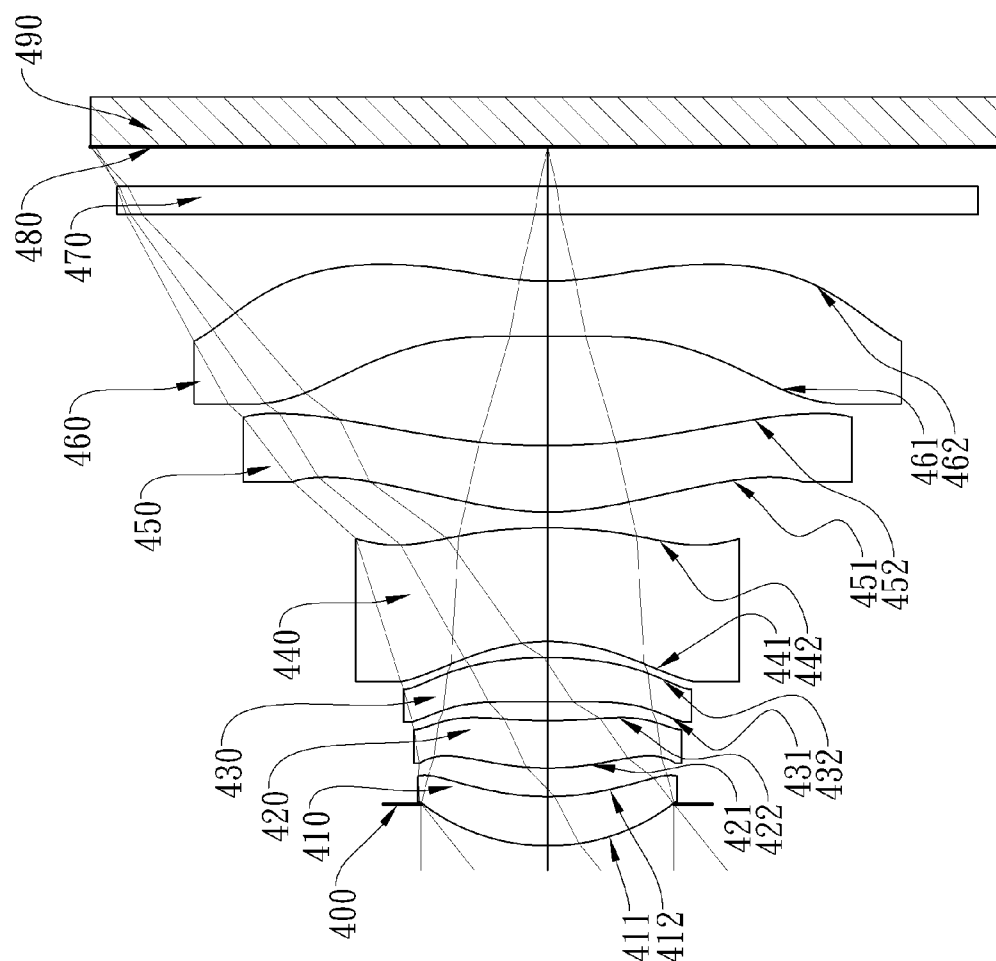


Fig. 4A

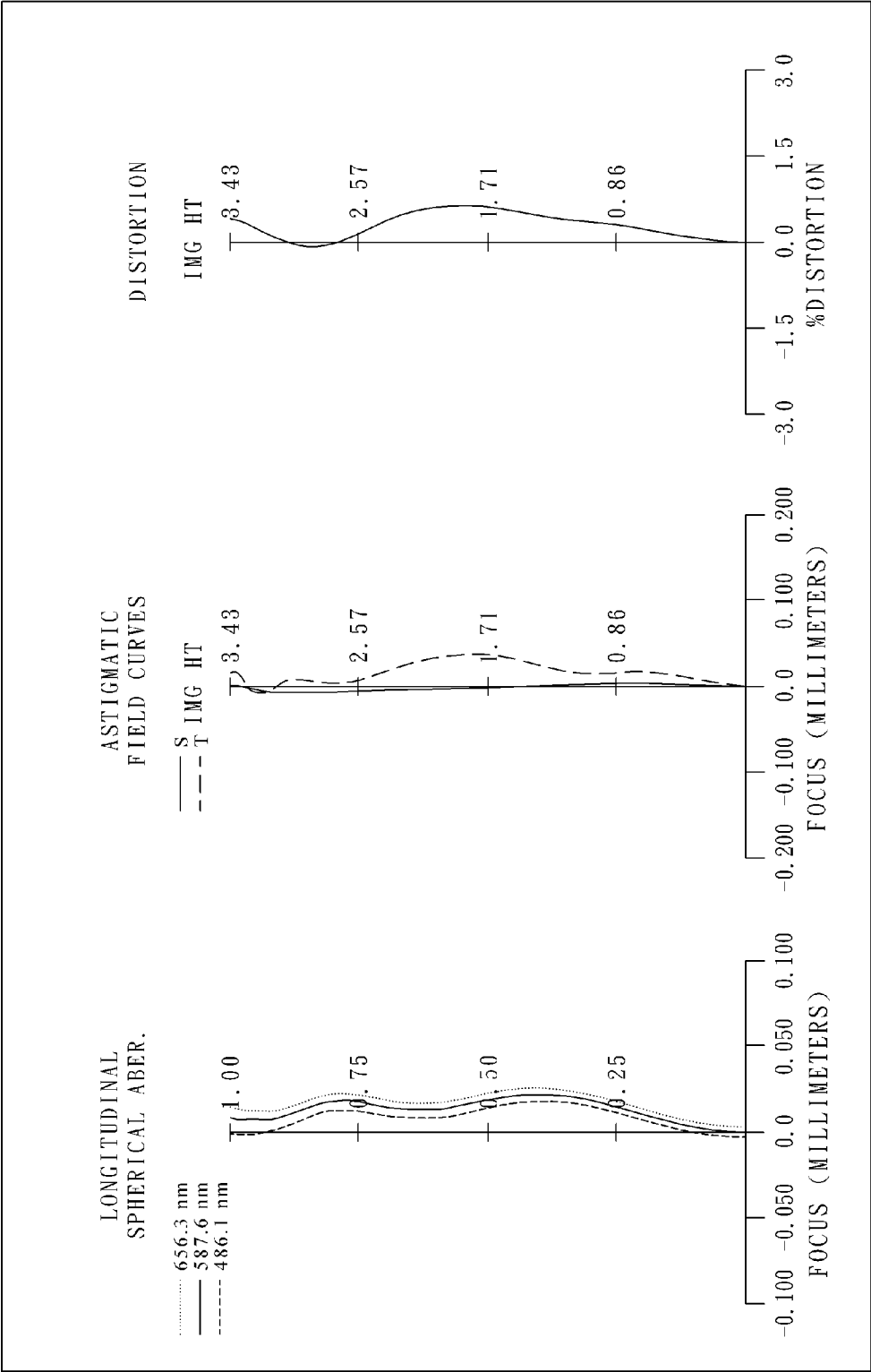


Fig. 4B

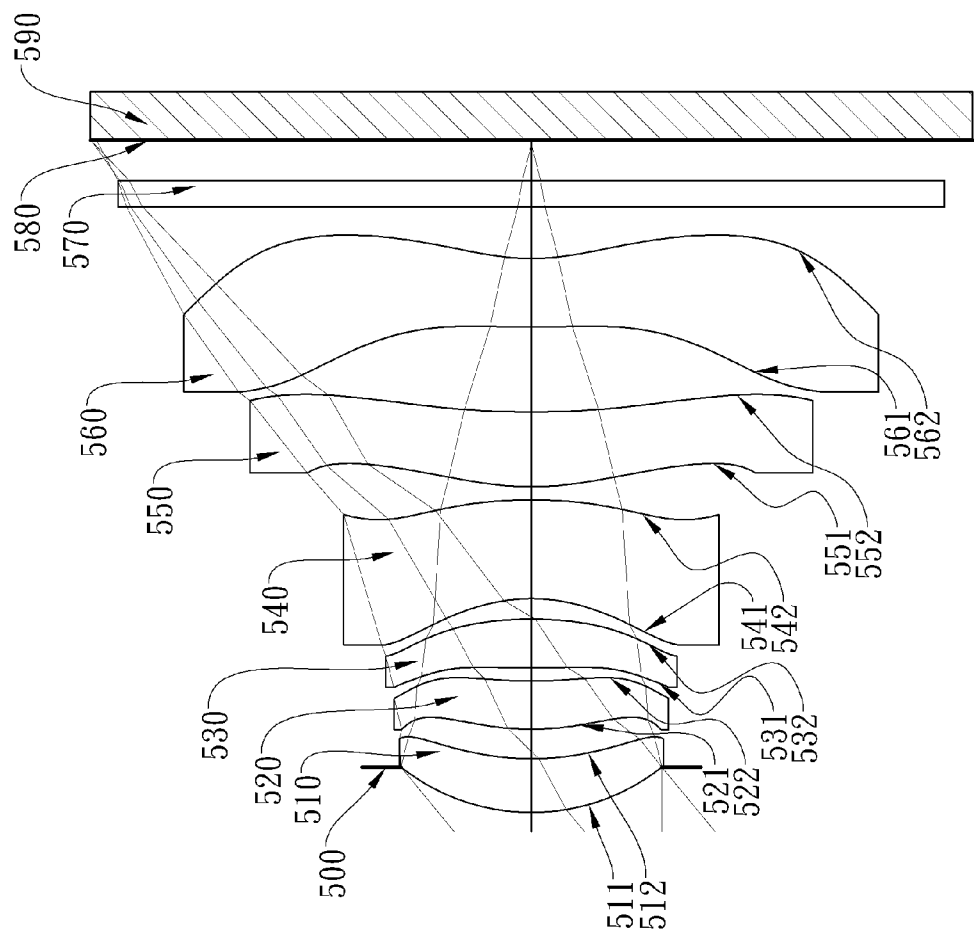


Fig. 5A

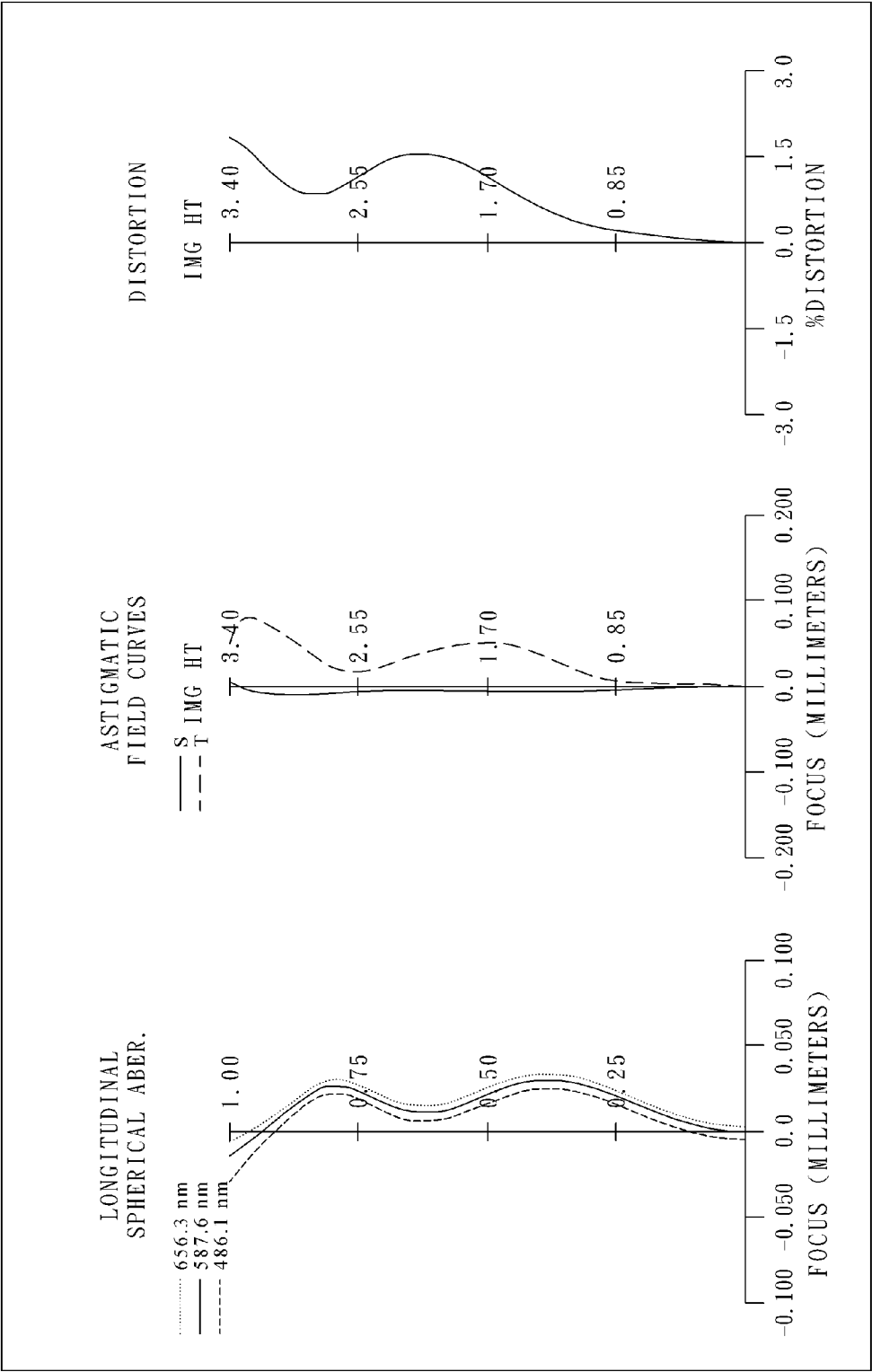


Fig. 5B

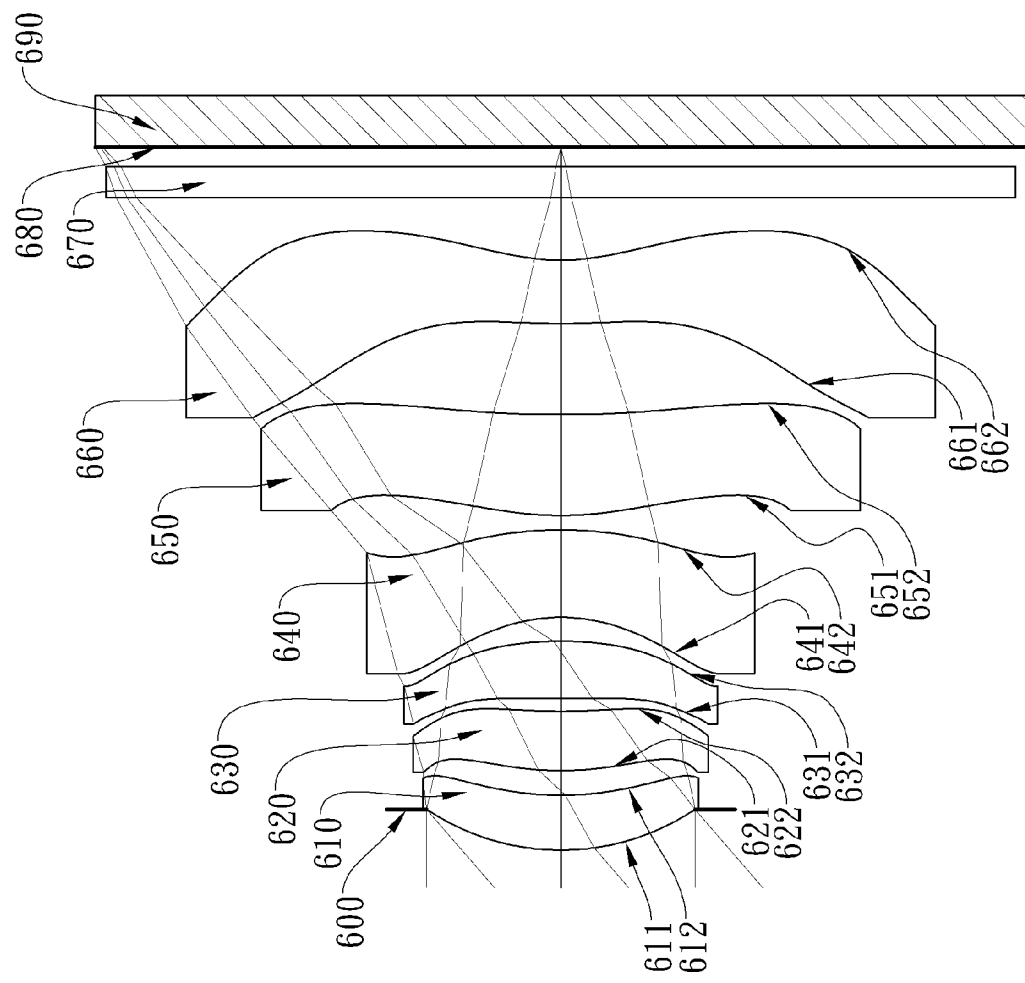


Fig. 6A

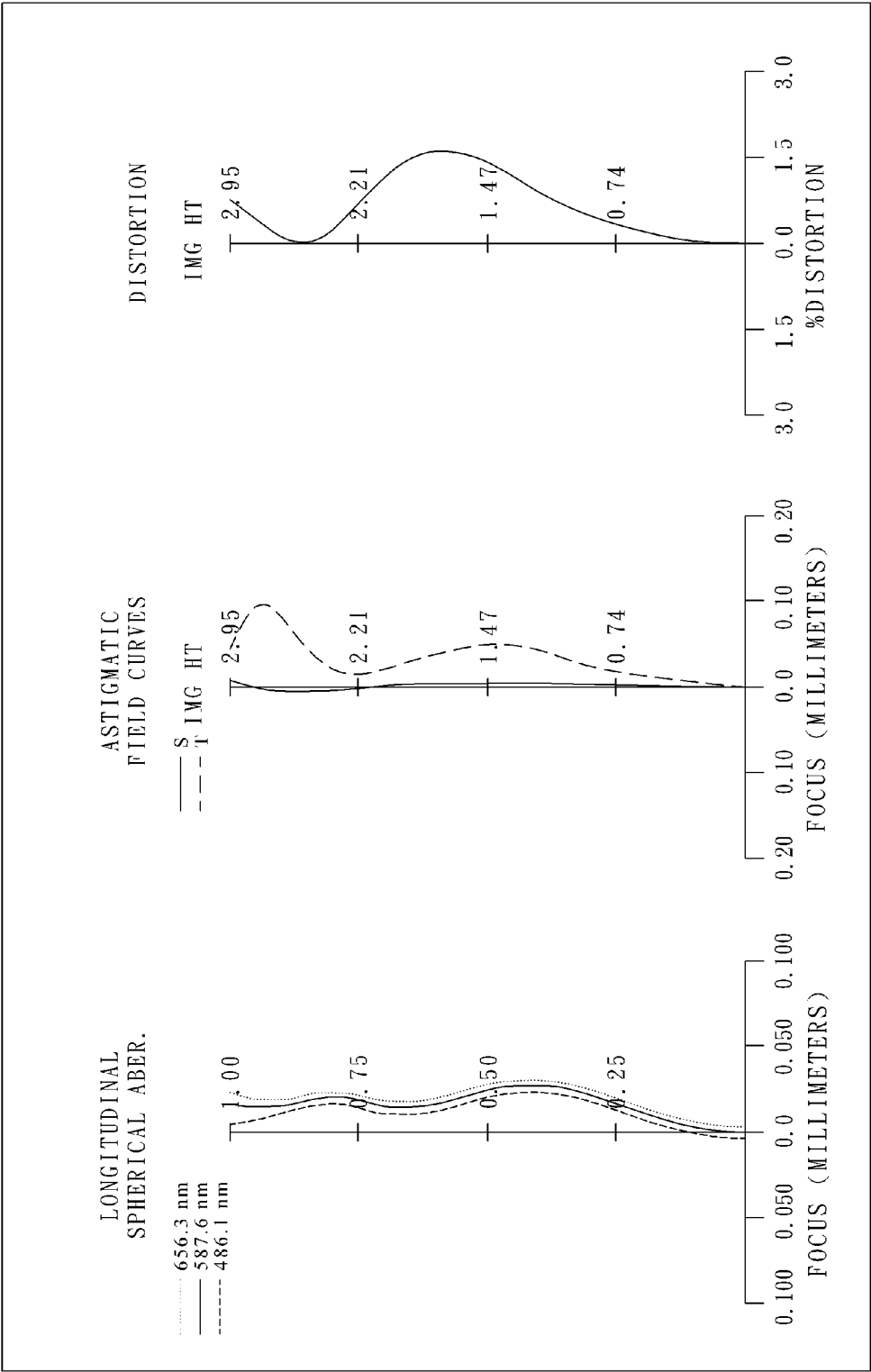


Fig. 6B

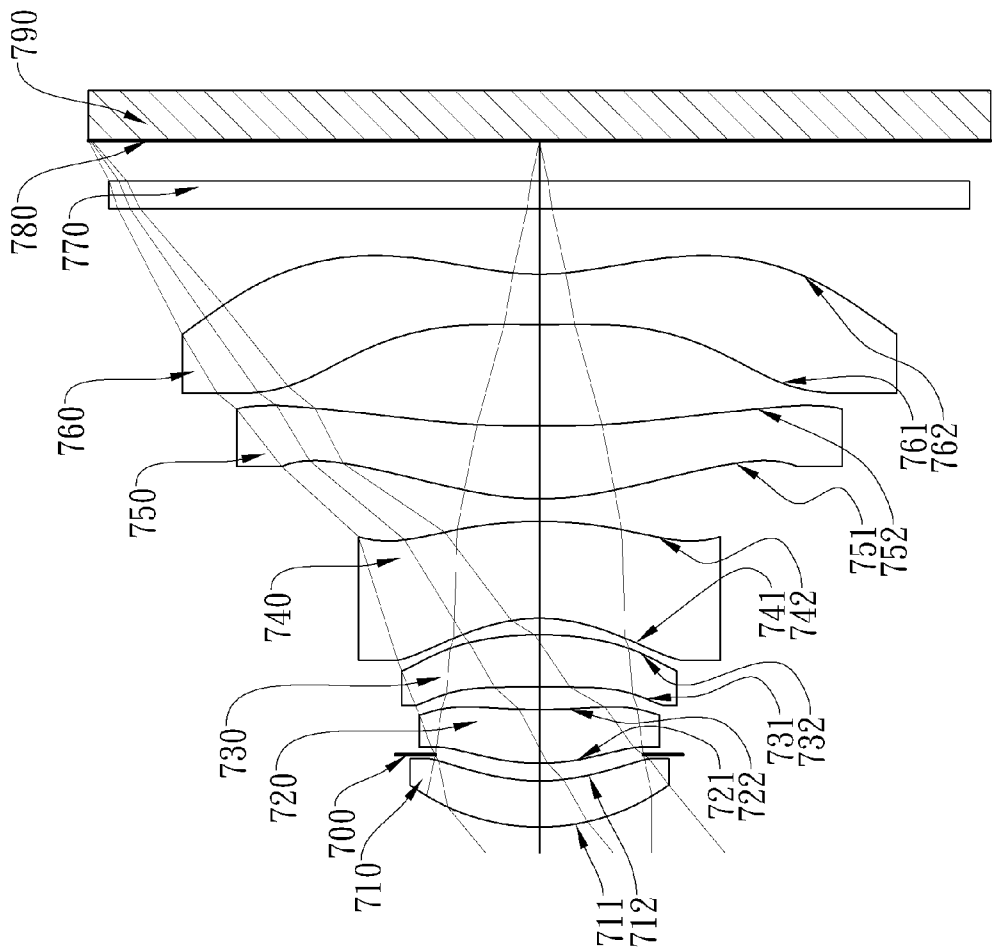


Fig. 7A

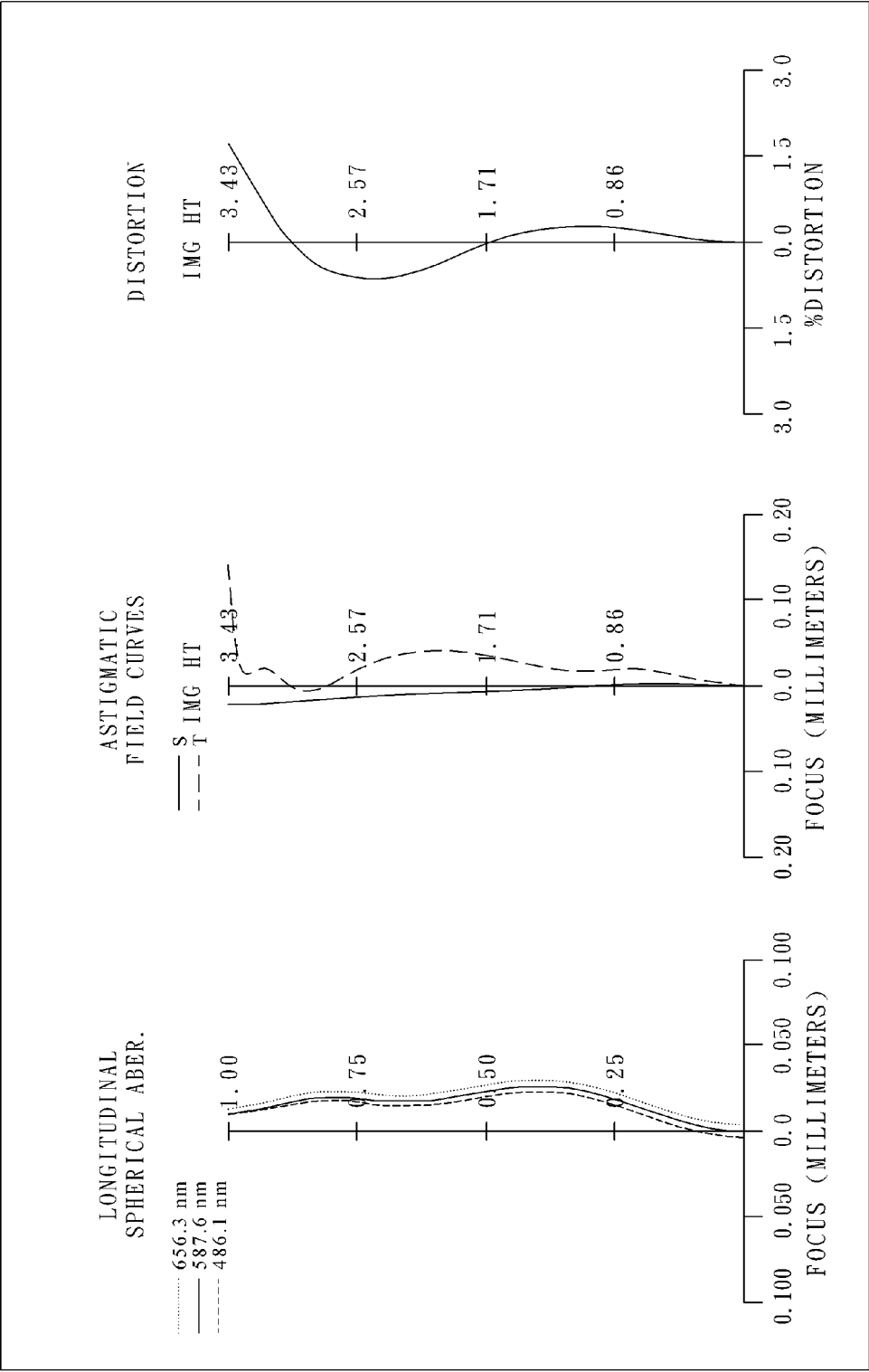


Fig. 7B

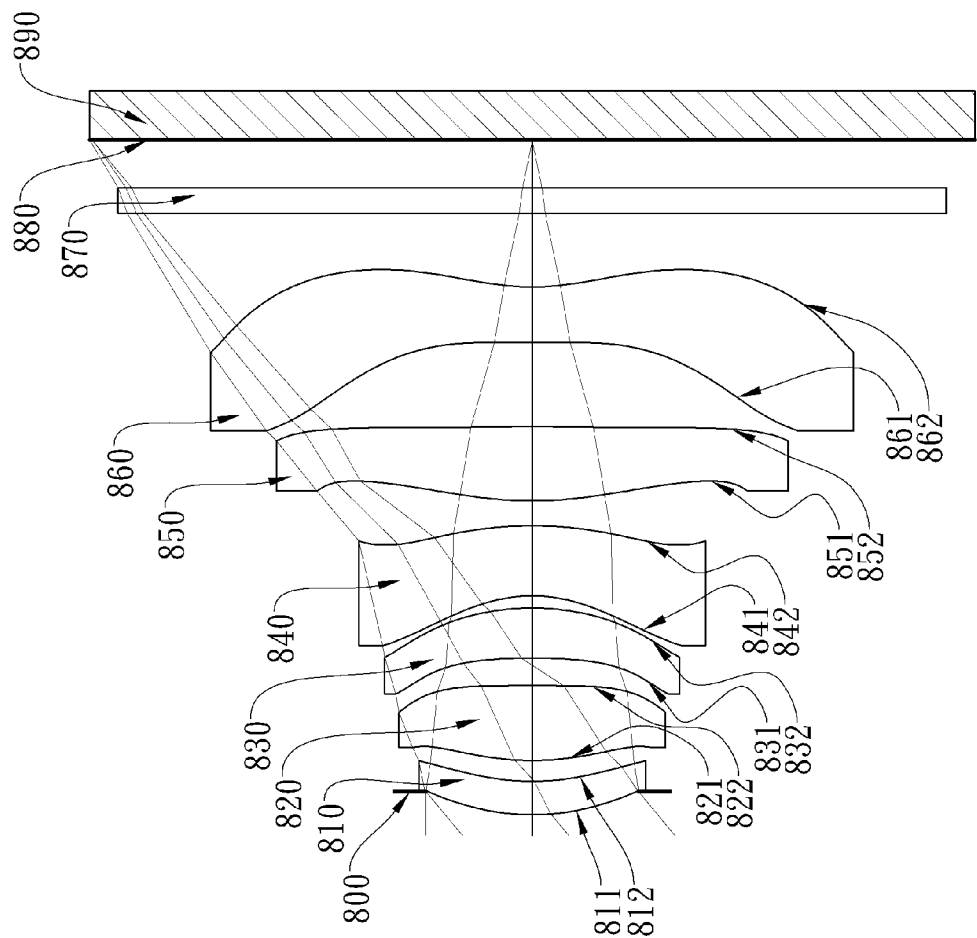


Fig. 8A

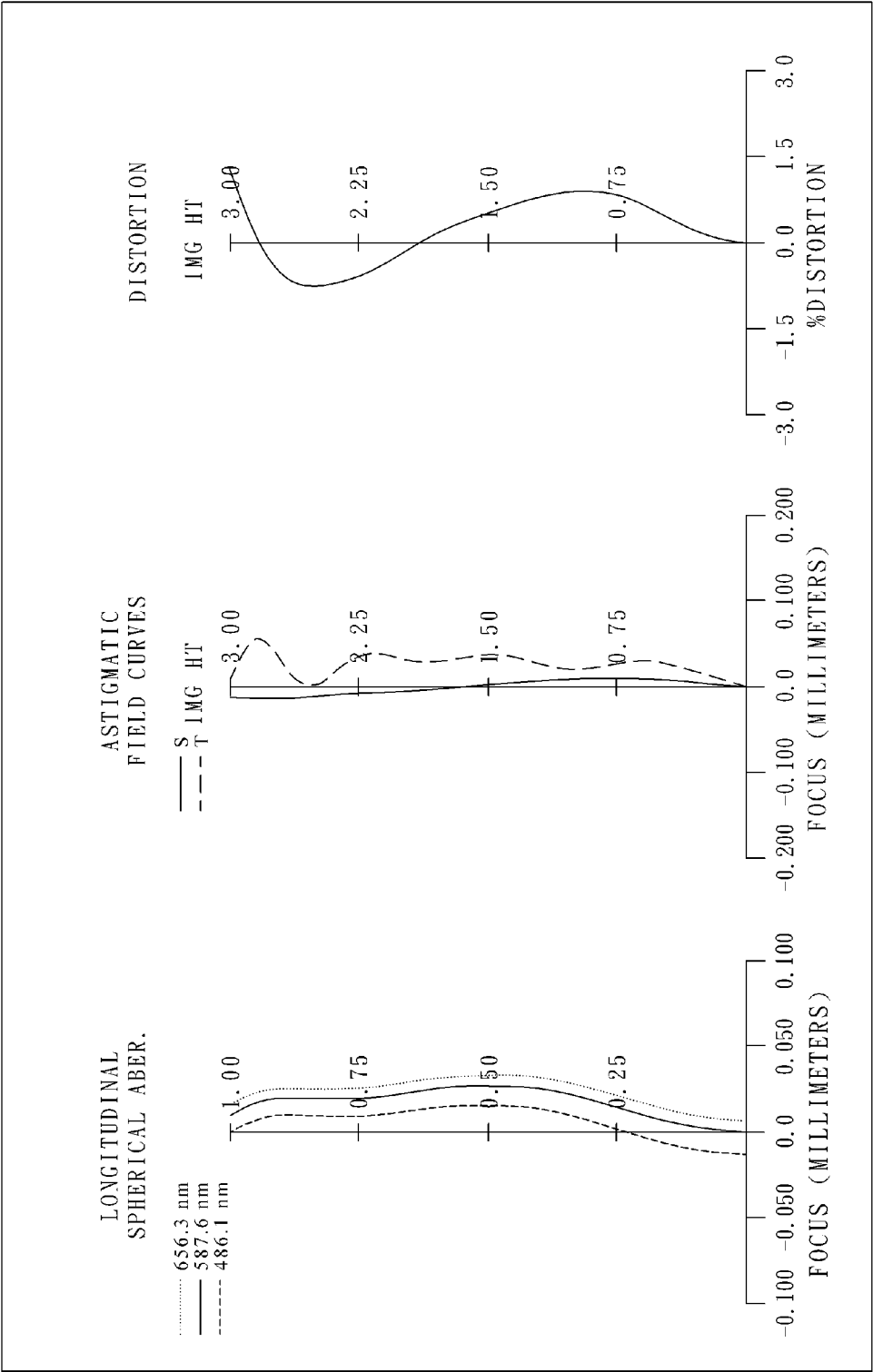


Fig. 8B

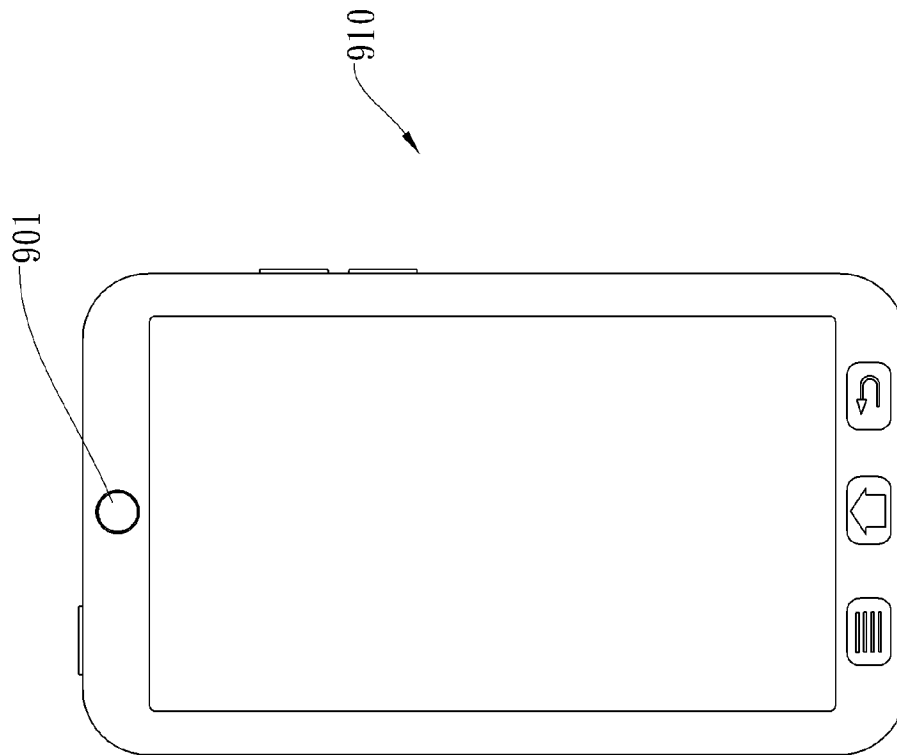


Fig. 9A

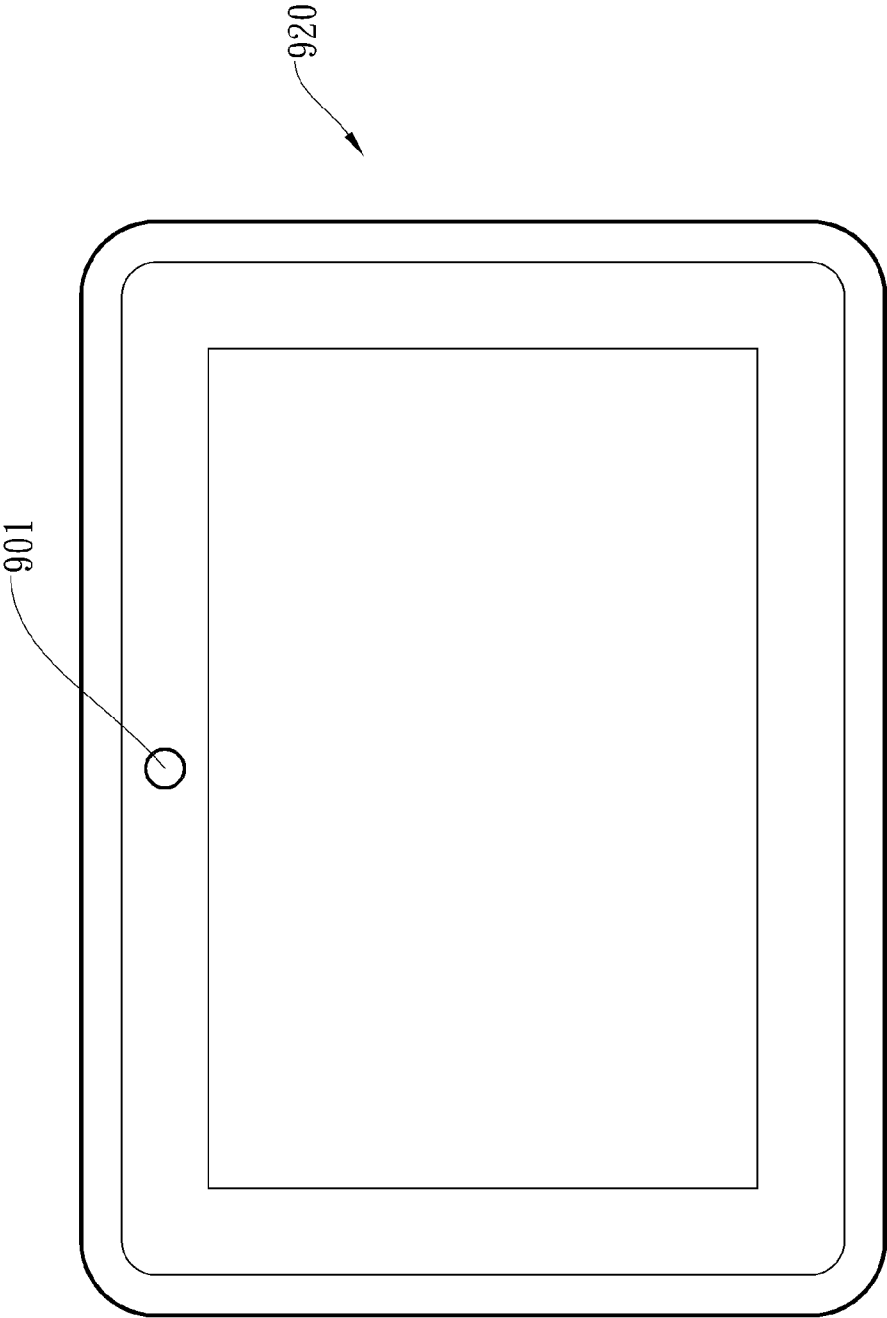


Fig. 9B

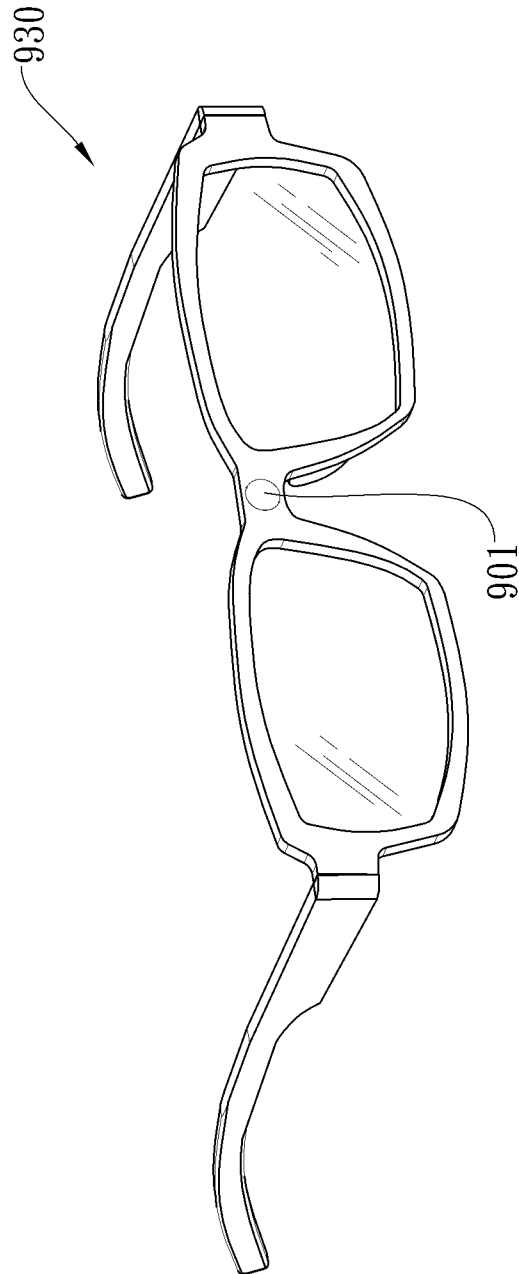


Fig. 9C

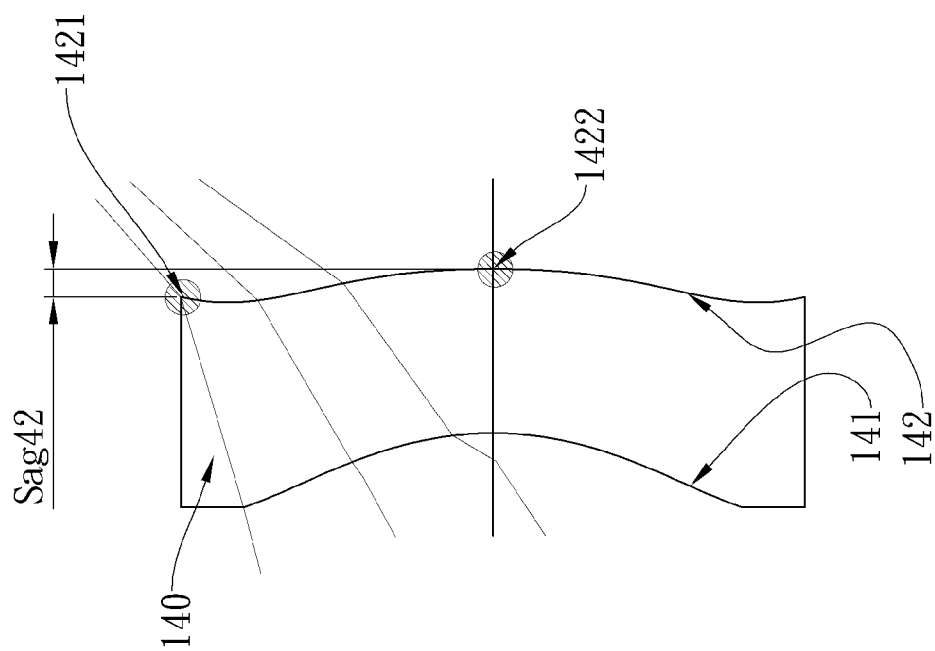


Fig. 10

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IMAGING LENS ASSEMBLY, IMAGING
DEVICE AND MOBILE TERMINAL

RELATED APPLICATIONS

This application claims priority to Taiwan Application Serial Number 103100925, filed Jan. 10, 2014, which is incorporated by reference herein in its entirety.

BACKGROUND

1. Technical Field

The present disclosure relates to an imaging lens assembly, an imaging device and a mobile terminal. More particularly, the present disclosure relates to a compact imaging lens assembly and an imaging device applicable to a mobile terminal.

2. Description of Related Art

In recent years, with the popularity of mobile terminals having camera functionalities, the demand of miniaturized optical systems has been increasing. The sensor of a conventional optical system is typically a CCD (Charge-Coupled Device) or a CMOS (Complementary Metal-Oxide-Semiconductor) sensor. As the advanced semiconductor manufacturing technologies have allowed the pixel size of sensors to be reduced and compact optical systems have gradually evolved toward the field of higher megapixels, there is an increasing demand for compact optical systems featuring better image quality.

A conventional optical system employed in a portable electronic product mainly adopts a five-element lens structure. Due to the popularity of mobile terminals with high-end specifications, such as smart phones, tablet personal computers and wearable apparatus, the requirements for high resolution and image quality of present compact optical systems increase significantly. However, the conventional optical systems cannot satisfy these requirements of the compact optical systems.

Other conventional compact optical systems with six-element lens structure enhance image quality and resolution. However, the arrangement of the refractive powers of the lens elements is not favorable for reducing the field curvature. Although the fourth lens element with negative refractive power is usually used to reduce the field curvature, the conventional curvature design of the fourth lens element is not favorable for having sufficient negative refractive power which results in being easily cracked or deformed.

Accordingly, a need exists in the art for an optical system that features better field curvature reduction and lens structure so as to have superior image quality.

SUMMARY

According to one aspect of the present disclosure, an imaging lens assembly includes, in order from an object side to an image side, a first lens element, a second lens element, a third lens element, a fourth lens element, a fifth lens element, and a sixth lens element. The first lens element has refractive power. The second lens element has refractive power. The third lens element has positive refractive power. The fourth lens element has negative refractive power. The fifth lens element with refractive power has an image-side surface being concave in a paraxial region thereof, wherein an object-side surface and the image-side surface of the fifth lens element are aspheric. The sixth lens element with refractive power has an image-side surface being concave in a paraxial region thereof, wherein the image-side surface of the sixth

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lens element has at least one convex shape in an off-axis region thereof, and an object-side surface and the image-side surface of the sixth lens element are aspheric. The imaging lens assembly has a total of six lens elements with refractive power. When a central thickness of the third lens element is $CT3$, a central thickness of the fourth lens element is $CT4$, a focal length of the imaging lens assembly is f , and a focal length of the third lens element is $f3$, the following conditions are satisfied:

$$1.40 < CT4/CT3; \text{ and}$$

$$0.35 < f/f3.$$

According to another aspect of the present disclosure, an imaging device includes the imaging lens assembly according to the aforementioned aspect and an image sensor.

According to still another aspect of the present disclosure, a mobile terminal includes the imaging device according to the aforementioned aspect.

According to yet another aspect of the present disclosure, an imaging lens assembly includes, in order from an object side to an image side, a first lens element, a second lens element, a third lens element, a fourth lens element, a fifth lens element, and a sixth lens element. The first lens element has refractive power. The second lens element has positive refractive power. The third lens element has positive refractive power. The fourth lens element has negative refractive power. The fifth lens element has refractive power, wherein an object-side surface and an image-side surface of the fifth lens element are aspheric. The sixth lens element with refractive power has an image-side surface being concave in a paraxial region thereof, wherein the image-side surface of the sixth lens element has at least one convex shape in an off-axis region thereof, and an object-side surface and the image-side surface of the sixth lens element are aspheric. The imaging lens assembly has a total of six lens elements with refractive power. When a central thickness of the third lens element is $CT3$, a central thickness of the fourth lens element is $CT4$, a focal length of the imaging lens assembly is f , and a focal length of the third lens element is $f3$, the following conditions are satisfied:

$$1.20 < CT4/CT3; \text{ and}$$

$$0.35 < f/f3.$$

According to still yet another aspect of the present disclosure, an imaging device includes the imaging lens assembly according to the foregoing aspect and an image sensor.

According to a further aspect of the present disclosure, a mobile terminal includes the imaging device according to the foregoing aspect.

According to a yet further aspect of the present disclosure, an imaging lens assembly includes, in order from an object side to an image side, a first lens element, a second lens element, a third lens element, a fourth lens element, a fifth lens element, and a sixth lens element. The first lens element has refractive power. The second lens element has refractive power. The third lens element has positive refractive power. The fourth lens element has negative refractive power. The fifth lens element has refractive power, wherein an object-side surface and an image-side surface of the fifth lens element are aspheric. The sixth lens element with refractive power has an image-side surface being concave in a paraxial region thereof, wherein the image-side surface of the sixth lens element has at least one convex shape in an off-axis region thereof, and an object-side surface and the image-side surface of the sixth lens element are aspheric. The imaging lens assembly has a

total of six lens elements with refractive power. When a central thickness of the third lens element is CT3, a central thickness of the fourth lens element is CT4, a focal length of the imaging lens assembly is f, and a focal length of the third lens element is f3, the following conditions are satisfied:

$$1.75 < CT4/CT3 < 3.50; \text{ and}$$

$$0.50 < f/f3.$$

BRIEF DESCRIPTION OF THE DRAWINGS

The disclosure can be more fully understood by reading the following detailed description of the embodiments, with reference made to the accompanying drawings as follows:

FIG. 1A is a schematic view of an imaging device according to the 1st embodiment of the present disclosure;

FIG. 1B shows spherical aberration curves, astigmatic field curves and a distortion curve of the imaging device according to the 1st embodiment;

FIG. 2A is a schematic view of an imaging device according to the 2nd embodiment of the present disclosure;

FIG. 2B shows spherical aberration curves, astigmatic field curves and a distortion curve of the imaging device according to the 2nd embodiment;

FIG. 3A is a schematic view of an imaging device according to the 3rd embodiment of the present disclosure;

FIG. 3B shows spherical aberration curves, astigmatic field curves and a distortion curve of the imaging device according to the 3rd embodiment;

FIG. 4A is a schematic view of an imaging device according to the 4th embodiment of the present disclosure;

FIG. 4B shows spherical aberration curves, astigmatic field curves and a distortion curve of the imaging device according to the 4th embodiment;

FIG. 5A is a schematic view of an imaging device according to the 5th embodiment of the present disclosure;

FIG. 5B shows spherical aberration curves, astigmatic field curves and a distortion curve of the imaging device according to the 5th embodiment;

FIG. 6A is a schematic view of an imaging device according to the 6th embodiment of the present disclosure;

FIG. 6B shows spherical aberration curves, astigmatic field curves and a distortion curve of the imaging device according to the 6th embodiment;

FIG. 7A is a schematic view of an imaging device according to the 7th embodiment of the present disclosure;

FIG. 7B shows spherical aberration curves, astigmatic field curves and a distortion curve of the imaging device according to the 7th embodiment;

FIG. 8A is a schematic view of an imaging device according to the 8th embodiment of the present disclosure;

FIG. 8B shows spherical aberration curves, astigmatic field curves and a distortion curve of the imaging device according to the 8th embodiment;

FIG. 9A shows a smart phone with an imaging device of the present disclosure installed therein;

FIG. 9B shows a tablet personal computer with an imaging device of the present disclosure installed therein;

FIG. 9C shows a wearable device with an imaging device of the present disclosure installed therein; and

FIG. 10 shows Sag42 of the present disclosure according to the FIG. 1.

DETAILED DESCRIPTION

An imaging lens assembly includes, in order from an object side to an image side, a first lens element, a second lens

element, a third lens element, a fourth lens element, a fifth lens element, and a sixth lens element. The imaging lens assembly has a total of six lens elements with refractive power.

The first lens element can have an object-side surface being convex in a paraxial region thereof and an image-side surface being concave in a paraxial region thereof, so that it is favorable for correcting aberration of the imaging lens assembly.

The second lens element can have positive refractive power. Therefore, it is favorable for balancing the positive refractive power of the third lens element. It is also favorable for reducing the photosensitivity and total track length of the imaging lens assembly. Moreover, the second lens element can have an object-side surface being convex in a paraxial region thereof. Therefore, it is favorable for correcting the astigmatism of the imaging lens assembly. In addition, the image-side surface of the second lens element can have at least one convex shape in an off-axis region thereof. Accordingly, it is favorable for reducing the incident angle of the off-axis on the image plane so as to improve the receiving efficiency of an image sensor and to further correct the aberration of the off-axis.

The third lens element has positive refractive power, so that it provides the imaging lens assembly with the positive refractive power as it needs to be so as to reduce the total track length. The third lens element can have an image-side surface being convex in a paraxial region thereof, so that it is favorable for correcting the spherical aberration of the imaging lens assembly.

The fourth lens element has negative refractive power, so that it is favorable for correcting the aberration of the third lens element with positive refractive power. Moreover, when the third lens element has positive refractive power and the fourth lens element has negative refractive power, a positive-negative telephoto structure is formed so that it is favorable for reducing the total track length of the imaging lens assembly.

The fifth lens element can have positive refractive power, so that it is favorable for reducing the spherical aberration and astigmatism and balancing the arrangement of the positive refractive powers. The fifth lens element can have an object-side surface being convex in a paraxial region thereof and an image-side surface being concave in a paraxial region thereof, so that it is favorable for effectively correcting the distortion and high order aberration of the peripheral region so as to improve the resolving power.

The sixth lens element can have negative refractive power, so that it is favorable for correcting the aberration of the imaging lens assembly. The sixth lens element has the image-side surface being concave in a paraxial region thereof, so that it is favorable for the principal point being positioned away from the image plane so as to reduce the total track length. Moreover, the image-side surface of the sixth lens element has at least one convex shape in an off-axis region thereof. Therefore, it is favorable for reducing the incident angle of the off-axis on the image plane so as to improve the receiving efficiency of the image sensor and to further correct the aberration of the off-axis.

When a focal length of the imaging lens assembly is f, and a focal length of the third lens element is f3, the following condition is satisfied: $0.35 < f/f3$. Therefore, it is favorable for balancing the arrangement of the refractive powers of the lens elements so as to reduce the total track length, maintain a compact size and high order spherical aberration. Preferably, the following condition is satisfied: $0.50 < f/f3$. More preferably, the following condition is satisfied: $0.60 < f/f3 < 1.50$.

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When a focal length of the second lens element is f_2 , and the focal length of the third lens element is f_3 , the following condition is satisfied: $0.75 < f_2/f_3 < 2.0$. Therefore, it is favorable for distributing the refractive powers of the lens elements so as to reduce the photosensitivity and aberration.

When a central thickness of the third lens element is CT_3 , and a central thickness of the fourth lens element is CT_4 , the following condition is satisfied: $1.20 < CT_4/CT_3$. Therefore, it is favorable for the fourth lens element having sufficient thickness so as to avoid being easily cracked and deformed and to further avoid reflectivity of having a larger angle. Preferably, the following condition is satisfied: $1.40 < CT_4/CT_3$. More preferably, the following condition is satisfied: $1.50 < CT_4/CT_3 < 4.0$. Further more preferably, the following condition is satisfied: $1.75 < CT_4/CT_3 < 3.50$.

When an axial distance between the first lens element and the second lens element is T_{12} , an axial distance between the second lens element and the third lens element is T_{23} , an axial distance between the third lens element and the fourth lens element is T_{34} , an axial distance between the fourth lens element and the fifth lens element is T_{45} , and an axial distance between the fifth lens element and the sixth lens element is T_{56} , the following condition is satisfied: $0.75 < T_{56}/(T_{12} + T_{23} + T_{34} + T_{45}) < 2.0$. Therefore, it is favorable for assembling the lens elements and improving the manufacturing yield rate.

When a distance in parallel with an optical axis from an axial vertex on the image-side surface of the fourth lens element to a maximum effective radius position on the image-side surface of the fourth lens element is Sag_{42} , and the central thickness of the fourth lens element is CT_4 , the following condition is satisfied: $|Sag_{42}|/CT_4 < 0.3$. Therefore, it is favorable for the surface curvature of the fourth lens element not being excessively curved and having a proper thickness. Moreover, it is favorable for manufacturing and molding the lens elements and keeping the lens elements more compact. When the distance towards the object side of the imaging lens assembly, Sag_{42} is negative; when the distance towards the image side of the imaging lens assembly, Sag_{42} is positive.

When an Abbe number of a lens element is V , only one lens element among all of the lens elements of the imaging lens assembly satisfies the following condition: $V < 30$. Therefore, it is favorable for effectively correcting the chromatic aberration of the imaging lens assembly.

When a curvature radius of the object-side surface of the first lens element is R_1 , and a curvature radius of the image-side surface of the first lens element is R_2 , the following condition is satisfied: $0.80 < R_2/R_1$. Therefore, it is favorable for the surface curvature of the first lens element so as to reduce the photosensitivity, total track length and correct the aberration.

When a focal length of the imaging lens assembly is f , and a curvature radius of the image-side surface of the sixth lens element is R_{12} , the following condition is satisfied: $0.20 < R_{12}/f < 0.75$. Therefore, it is favorable for further correcting the high order aberration. Preferably, the following condition is satisfied: $0.30 < R_{12}/f < 0.75$.

When a focal length of the first lens element is f_1 , the focal length of the second lens element is f_2 , and the focal length of the third lens element is f_3 , the following condition is satisfied: $|f_3/f_1| + |f_3/f_2| < 1.5$. Therefore, it is favorable for arranging the refractive powers of the imaging lens assembly so as to reduce spherical aberration, astigmatism and aberration.

According to the imaging lens assembly of the present disclosure, the lens elements thereof can be made of glass or plastic material. When the lens elements are made of glass

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material, the distribution of the refractive power of the imaging lens assembly may be more flexible to design. When the lens elements are made of plastic material, the manufacturing cost can be effectively reduced. Furthermore, surfaces of each lens element can be arranged to be aspheric, since the aspheric surface of the lens element is easy to form a shape other than spherical surface so as to have more controllable variables for eliminating the aberration thereof, and to further decrease the required number of the lens elements. Therefore, the total track length of the imaging lens assembly can also be reduced.

According to the imaging lens assembly of the present disclosure, each of an object-side surface and an image-side surface has a paraxial region and an off-axis region. The paraxial region refers to the region of the surface where light rays travel close to the optical axis, and the off-axis region refers to the region of the surface where light rays travel away from the optical axis. Particularly, when the lens element has a convex surface, it indicates that the surface is convex in the paraxial region thereof; when the lens element has a concave surface, it indicates that the surface is concave in the paraxial region thereof.

According to the imaging lens assembly of the present disclosure, the imaging lens assembly can include at least one stop, such as an aperture stop, a glare stop or a field stop. Said glare stop or said field stop is for eliminating the stray light and thereby improving the image resolution thereof.

According to the imaging lens assembly of the present disclosure, an aperture stop can be configured as a front stop or a middle stop. A front stop disposed between an imaged object and the first lens element can provide a longer distance between an exit pupil of the imaging lens assembly and the image plane and thereby improves the image-sensing efficiency of an image sensor. A middle stop disposed between the first lens element and the image plane is favorable for enlarging the field of view of the imaging lens assembly and thereby provides a wider field of view for the same.

The present imaging lens assembly can be optionally applied to moving focus optical systems. According to the imaging lens assembly of the present disclosure, the imaging lens assembly is featured with good correction ability and high image quality, and can be applied to 3D (three-dimensional) image capturing applications, in products such as digital cameras, mobile devices, digital tablets, wearable devices and other mobile terminals.

According to the present disclosure, an imaging device is provided. The imaging device includes the imaging lens assembly according to the aforementioned imaging lens assembly of the present disclosure, and an image sensor, wherein the image sensor is disposed on an image plane of the aforementioned imaging lens assembly. In some embodiments, the imaging device can further include a barrel member, a holding member or a combination thereof.

In FIG. 9A, FIG. 9B and FIG. 9C, an imaging device 901 may be installed in but not limited to a mobile terminal, including a smart phone (910), a tablet personal computer (920) or a wearable device (930). The three exemplary figures of different kinds of mobile terminal are only exemplary for showing the imaging device of present disclosure installing in a mobile terminal and is not limited thereto. In some embodiments, the mobile terminal can further include, but not limited to, display, a control unit, a random access memory unit (RAM), a read only memory unit (ROM) or a combination thereof.

According to the above description of the present disclosure, the following 1st-8th specific embodiments are provided for further explanation.

FIG. 1A is a schematic view of an imaging device according to the 1st embodiment of the present disclosure. FIG. 1B shows, in order from left to right, spherical aberration curves, astigmatic field curves and a distortion curve of the imaging device according to the 1st embodiment. In FIG. 1A, the imaging device includes the imaging lens assembly (not otherwise herein labeled) of the present disclosure and an image sensor 190. The imaging lens assembly includes, in order from an object side to an image side, an aperture stop 100, a first lens element 110, a second lens element 120, a third lens element 130, a fourth lens element 140, a fifth lens element 150, a sixth lens element 160, an IR-cut filter 170 and an image plane 180, wherein the imaging lens assembly has a total of six lens elements (110-160) with refractive power.

The first lens element 110 with positive refractive power has an object-side surface 111 being convex in a paraxial region thereof and an image-side surface 112 being concave in a paraxial region thereof. The first lens element 110 is made of plastic material and has the object-side surface 111 and the image-side surface 112 being both aspheric.

The second lens element 120 with positive refractive power has an object-side surface 121 being convex in a paraxial region thereof and an image-side surface 122 being concave in a paraxial region thereof, wherein the image-side surface 122 of the second lens element 120 has at least one convex shape in an off-axis region thereof. The second lens element 120 is made of plastic material and has the object-side surface 121 and the image-side surface 122 being both aspheric.

The third lens element 130 with positive refractive power has an object-side surface 131 being concave in a paraxial region thereof and an image-side surface 132 being convex in a paraxial region thereof. The third lens element 130 is made of plastic material and has the object-side surface 131 and the image-side surface 132 being both aspheric.

The fourth lens element 140 with negative refractive power has an object-side surface 141 being concave in a paraxial region thereof and an image-side surface 142 being convex in a paraxial region thereof. The fourth lens element 140 is made of plastic material and has the object-side surface 141 and the image-side surface 142 being both aspheric.

The fifth lens element 150 with positive refractive power has an object-side surface 151 being convex in a paraxial region thereof and an image-side surface 152 being concave in a paraxial region thereof. The fifth lens element 150 is made of plastic material and has the object-side surface 151 and the image-side surface 152 being both aspheric.

The sixth lens element 160 with negative refractive power has an object-side surface 161 being convex in a paraxial region thereof and an image-side surface 162 being concave in a paraxial region thereof, wherein the image-side surface 162 of the sixth lens element 160 has at least one convex shape in an off-axis region thereof. The sixth lens element 160 is made of plastic material and has the object-side surface 161 and the image-side surface 162 being both aspheric.

The IR-cut filter 170 is made of glass and located between the sixth lens element 160 and the image plane 180, and will not affect the focal length of the imaging lens assembly. The image sensor 190 is disposed on the image plane 180 of the imaging lens assembly.

The equation of the aspheric surface profiles of the aforementioned lens elements of the 1st embodiment is expressed as follows:

$$X(Y) = (Y^2 / R) / (1 + \sqrt{1 - (1 + k) \times (Y / R)^2}) + \sum_i (Ai) \times (Y^i),$$

where,

X is the relative distance between a point on the aspheric surface spaced at a distance Y from the optical axis and the tangential plane at the aspheric surface vertex on the optical axis;

Y is the vertical distance from the point on the aspheric surface to the optical axis;

R is the curvature radius;

k is the conic coefficient; and

Ai is the i-th aspheric coefficient, and in the embodiments, i may be, but is not limited to, 4, 6, 8, 10, 12, 14 and 16.

In the imaging lens assembly of the imaging device according to the 1st embodiment, when a focal length of the imaging lens assembly is f, an f-number of the imaging lens assembly is Fno, and half of a maximal field of view of the imaging lens assembly is HFOV, these parameters have the following values: f=4.42 mm; Fno=2.00; and HFOV=40.0 degrees.

In the imaging lens assembly of the imaging device according to the 1st embodiment, when a central thickness of the third lens element 130 is CT3, and a central thickness of the fourth lens element 140 is CT4, the following condition is satisfied: CT4/CT3=2.20.

In the imaging lens assembly of the imaging device according to the 1st embodiment, when an axial distance between the first lens element 110 and the second lens element 120 is T12, an axial distance between the second lens element 120 and the third lens element 130 is T23, an axial distance between the third lens element 130 and the fourth lens element 140 is T34, an axial distance between the fourth lens element 140 and the fifth lens element 150 is T45, and an axial distance between the fifth lens element 150 and the sixth lens element 160 is T56, the following condition is satisfied: T56/(T12+T23+T34+T45)=1.20.

In FIG. 10, when a distance in parallel with an optical axis from an axial vertex 1422 on the image-side surface 142 of the fourth lens element 140 to a maximum effective radius position 1421 on the image-side surface 142 of the fourth lens element 140 is Sag42, and the central thickness of the fourth lens element 140 is CT4, the following condition is satisfied: |Sag42|/CT4=0.17. When the distance towards the object side of the imaging lens assembly, Sag42 is negative; when the distance towards the image side of the imaging lens assembly, Sag42 is positive.

In the imaging lens assembly of the imaging device according to the 1st embodiment, when a curvature radius of the object-side surface 111 of the first lens element 110 is R1, and a curvature radius of the image-side surface 112 of the first lens element 110 is R2, the following condition is satisfied: |R2|/R1=1.23.

In the imaging lens assembly of the imaging device according to the 1st embodiment, when the focal length of the imaging lens assembly is f, and a curvature radius of the image-side surface 162 of the sixth lens element 160 is R12, the following condition is satisfied: R12/f=0.43.

In the imaging lens assembly of the imaging device according to the 1st embodiment, when the focal length of the

imaging lens assembly is f , and a focal length of the third lens element **130** is f_3 , the following condition is satisfied: $f/f_3=0.79$.

In the imaging lens assembly of the imaging device according to the 1st embodiment, when a focal length of the second lens element **120** is f_2 , and the focal length of the third lens element **130** is f_3 , the following condition is satisfied: $f_2/f_3=1.27$.

In the imaging lens assembly of the imaging device according to the 1st embodiment, when a focal length of the first lens element **110** is f_1 , the focal length of the second lens element **120** is f_2 , and the focal length of the third lens element **130** is f_3 , the following condition is satisfied: $|f_3/f_1|+|f_3/f_2|=1.21$.

The detailed optical data of the 1st embodiment are shown in Table 1 and the aspheric surface data are shown in Table 2 below.

TABLE 1

1st Embodiment							
$f = 4.42 \text{ mm}$, $F_{\text{no}} = 2.00$, $\text{HFOV} = 40.0 \text{ deg.}$							
Surface #		Curvature Radius	Thickness	Material	Index	Abbe #	Focal Length
0	Object	Plano	Infinity				
1	Ape. Stop	Plano	-0.367				
2	Lens 1	1.817 ASP	0.429	Plastic	1.544	55.9	13.09
3		2.236 ASP	0.232				
4	Lens 2	2.147 ASP	0.390	Plastic	1.544	55.9	7.11
5		4.512 ASP	0.150				
6	Lens 3	-63.436 ASP	0.378	Plastic	1.544	55.9	5.58
7		-2.905 ASP	0.142				
8	Lens 4	-1.633 ASP	0.832	Plastic	1.650	21.4	-4.22
9		-4.852 ASP	0.198				
10	Lens 5	2.076 ASP	0.592	Plastic	1.544	55.9	6.07
11		5.024 ASP	0.865				
12	Lens 6	7.771 ASP	0.522	Plastic	1.544	55.9	-4.83
13		1.917 ASP	0.500				
14	IR-cut filter	Plano	0.200	Glass	1.517	64.2	—
15		Plano	0.241				
16	Image	Plano	—				

Note:

Reference wavelength is 587.6 nm (d-line).

TABLE 2

Aspheric Coefficients						
Surface #	2	3	4	5	6	7
k =	7.5067E-01	1.5034E+00	-6.8711E+00	-2.1669E+01	-5.0000E+01	-7.7624E+00
A4 =	-2.4523E-02	-8.3677E-02	-1.9832E-02	-4.9049E-02	-6.2330E-02	-3.5123E-02
A6 =	-3.1171E-02	2.0068E-02	-3.7169E-02	-1.7929E-01	-7.3987E-02	-6.7515E-02
A8 =	1.3719E-01	1.6785E-02	-7.9339E-02	1.6379E-01	-1.2189E-01	-5.4575E-03
A10 =	-2.4171E-01	-8.0961E-02	1.4081E-01	-2.2352E-01	1.8740E-01	1.3160E-01
A12 =	1.9045E-01	7.9897E-02	-1.5160E-01	2.0092E-01	-2.0524E-02	-1.1427E-01
A14 =	-5.9210E-02	-3.9000E-02	5.0291E-02	-5.7173E-02	-2.3335E-02	3.1067E-02
Surface #	8	9	10	11	12	13
k =	-7.1219E-01	3.3014E+00	-1.0927E+01	-6.5741E+00	-2.0538E+01	-7.2425E+00
A4 =	6.8397E-02	-8.2582E-02	5.9095E-03	2.3111E-02	-1.4548E-01	-7.3396E-02
A6 =	-1.0874E-01	1.1206E-01	-1.8138E-02	-2.6683E-02	7.2970E-02	3.2435E-02
A8 =	1.7993E-01	-7.5939E-02	8.9963E-03	1.2019E-02	-2.6554E-02	-1.0178E-02
A10 =	-1.4770E-01	4.0333E-02	-2.2898E-03	-3.0187E-03	6.2848E-03	2.0360E-03
A12 =	6.6090E-02	-1.1646E-02	1.6464E-04	4.1385E-04	-8.5650E-04	-2.5332E-04
A14 =	-1.2411E-02	1.2966E-03	2.8910E-05	-2.7995E-05	6.1408E-05	1.7448E-05
A16 =			-5.0633E-06	6.8502E-07	-1.8165E-06	-4.9920E-07

In Table 1, the curvature radius, the thickness and the focal length are shown in millimeters (mm). Surface numbers 0-16 represent the surfaces sequentially arranged from the object-side to the image-side along the optical axis. In Table 2, k

represents the conic coefficient of the equation of the aspheric surface profiles. A4-A16 represent the aspheric coefficients ranging from the 4th order to the 16th order. The tables presented below for each embodiment are the corresponding schematic parameter and aberration curves, and the definitions of the tables are the same as Table 1 and Table 2 of the 1st embodiment. Therefore, an explanation in this regard will not be provided again.

2nd Embodiment

FIG. 2A is a schematic view of an imaging device according to the 2nd embodiment of the present disclosure. FIG. 2B shows, in order from left to right, spherical aberration curves, astigmatic field curves and a distortion curve of the imaging device according to the 2nd embodiment. In FIG. 2A, the

imaging device includes the imaging lens assembly (not otherwise herein labeled) of the present disclosure and an image sensor **290**. The imaging lens assembly includes, in order from an object side to an image side, an aperture stop **200**, a

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first lens element **210**, a second lens element **220**, a third lens element **230**, a fourth lens element **240**, a stop **202**, a fifth lens element **250**, a sixth lens element **260**, an IR-cut filter **270** and an image plane **280**, wherein the imaging lens assembly has a total of six lens elements (**210-260**) with refractive power.

The first lens element **210** with positive refractive power has an object-side surface **211** being convex in a paraxial region thereof and an image-side surface **212** being concave in a paraxial region thereof. The first lens element **210** is made of plastic material and has the object-side surface **211** and the image-side surface **212** being both aspheric.

The second lens element **220** with positive refractive power has an object-side surface **221** being convex in a paraxial region thereof and an image-side surface **222** being concave in a paraxial region thereof, wherein the image-side surface **222** of the second lens element **220** has at least one convex shape in an off-axis region thereof. The second lens element **220** is made of plastic material and has the object-side surface **221** and the image-side surface **222** being both aspheric.

The third lens element **230** with positive refractive power has an object-side surface **231** being concave in a paraxial region thereof and an image-side surface **232** being convex in a paraxial region thereof. The third lens element **230** is made of plastic material and has the object-side surface **231** and the image-side surface **232** being both aspheric.

The fourth lens element **240** with negative refractive power has an object-side surface **241** being concave in a paraxial

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region thereof and an image-side surface **242** being convex in a paraxial region thereof. The fourth lens element **240** is made of plastic material and has the object-side surface **241** and the image-side surface **242** being both aspheric.

The fifth lens element **250** with positive refractive power has an object-side surface **251** being convex in a paraxial region thereof and an image-side surface **252** being concave in a paraxial region thereof. The fifth lens element **250** is made of plastic material and has the object-side surface **251** and the image-side surface **252** being both aspheric.

The sixth lens element **260** with negative refractive power has an object-side surface **261** being concave in a paraxial region thereof and an image-side surface **262** being concave in a paraxial region thereof, wherein the image-side surface **262** of the sixth lens element **260** has at least one convex shape in an off-axis region thereof. The sixth lens element **260** is made of plastic material and has the object-side surface **261** and the image-side surface **262** being both aspheric.

The IR-cut filter **270** is made of glass and located between the sixth lens element **260** and the image plane **280**, and will not affect the focal length of the imaging lens assembly. The image sensor **290** is disposed on the image plane **280** of the imaging lens assembly.

The detailed optical data of the 2nd embodiment are shown in Table 3 and the aspheric surface data are shown in Table 4 below.

TABLE 3

2nd Embodiment							
f = 4.56 mm, Fno = 2.00, HFOV = 39.0 deg.							
Surface #		Curvature Radius	Thickness	Material	Index	Abbe #	Focal Length
0	Object	Plano	Infinity				
1	Ape. Stop	Plano	-0.389				
2	Lens 1	1.826 ASP	0.405	Plastic	1.570	55.0	15.66
3		2.111 ASP	0.265				
4	Lens 2	2.110 ASP	0.426	Plastic	1.570	55.0	6.11
5		4.965 ASP	0.147				
6	Lens 3	-42.126 ASP	0.337	Plastic	1.570	55.0	5.69
7		-3.022 ASP	0.139				
8	Lens 4	-1.684 ASP	0.950	Plastic	1.634	23.8	-4.13
9		-5.737 ASP	0.100				
10	Stop	Plano	0.108				
11	Lens 5	2.332 ASP	0.702	Plastic	1.570	55.0	6.92
12		5.083 ASP	1.088				
13	Lens 6	-88.865 ASP	0.380	Plastic	1.570	55.0	-4.94
14		2.910 ASP	0.400				
15	IR-cut filter	Plano	0.200	Glass	1.517	64.2	—
16		Plano	0.170				
17	Image	Plano	—				

Note:

Reference wavelength is 587.6 nm (d-line).

The effective radius of Surface 10 is 1.770 mm.

TABLE 4

Aspheric Coefficients						
Surface #	2	3	4	5	6	7
k =	1.0199E+00	1.7620E+00	-5.7477E+00	-2.2540E+01	-1.0000E+00	-6.9286E+00
A4 =	-3.1060E-02	-8.7636E-02	-1.1602E-02	-3.9747E-02	-6.0692E-02	-4.2016E-02
A6 =	-3.9546E-02	1.4706E-02	-3.0102E-02	-1.6955E-01	-6.0499E-02	-6.8799E-02
A8 =	1.4344E-01	1.4800E-02	-7.9607E-02	1.6231E-01	-1.2476E-01	-6.6592E-04
A10 =	-2.4413E-01	-8.1791E-02	1.3782E-01	-2.3068E-01	1.8222E-01	1.3482E-01
A12 =	1.8452E-01	8.0549E-02	-1.4968E-01	1.9594E-01	-2.3536E-02	-1.1294E-01
A14 =	-5.6147E-02	-3.6884E-02	5.0279E-02	-5.5149E-02	-1.9976E-02	2.9523E-02

TABLE 4-continued

Aspheric Coefficients						
Surface #	8	9	11	12	13	14
k =	-6.2799E-01	-1.3294E+00	-1.5541E+01	-5.0000E+01	-1.0000E+00	-1.5485E+01
A4 =	6.2454E-02	-7.3244E-02	-2.0967E-03	2.2419E-02	-1.3891E-01	-7.4066E-02
A6 =	-1.0901E-01	1.0907E-01	-1.5486E-02	-2.6815E-02	7.1585E-02	3.2588E-02
A8 =	1.8206E-01	-7.7046E-02	9.2261E-03	1.2095E-02	-2.6643E-02	-1.0300E-02
A10 =	-1.4656E-01	4.0508E-02	-2.3923E-03	-3.0051E-03	6.2698E-03	2.0420E-03
A12 =	6.5998E-02	-1.1477E-02	1.6788E-04	4.1043E-04	-8.5696E-04	-2.5231E-04
A14 =	-1.2241E-02	1.2498E-03	3.4689E-05	-2.8459E-05	6.1605E-05	1.7453E-05
A16 =			-5.2772E-06	7.0816E-07	-1.7766E-06	-5.0544E-07

In the 2nd embodiment, the equation of the aspheric surface profiles of the aforementioned lens elements is the same as the equation of the 1st embodiment. Also, the definitions of these parameters shown in the following table are the same as those stated in the 1st embodiment with corresponding values for the 2nd embodiment, so an explanation in this regard will not be provided again.

Moreover, these parameters can be calculated from Table 3 and Table 4 as the following values and satisfy the following conditions:

2nd Embodiment			
f [mm]	4.56	R2 /R1	1.16
Fno	2.00	R12/f	0.64
HFOV [deg.]	39.0	f/f3	0.80
CT4/CT3	2.82	f2/f3	1.07
T56/(T12 + T23 + T34 + T45)	1.43	f3/f1 + f3/f2	1.29
Sag42 /CT4	0.06		

3rd Embodiment

FIG. 3A is a schematic view of an imaging device according to the 3rd embodiment of the present disclosure. FIG. 3B shows, in order from left to right, spherical aberration curves, astigmatic field curves and a distortion curve of the imaging device according to the 3rd embodiment. In FIG. 3A, the imaging device includes the imaging lens assembly (not otherwise herein labeled) of the present disclosure and an image sensor 390. The imaging lens assembly includes, in order from an object side to an image side, an aperture stop 300, a first lens element 310, a second lens element 320, a third lens element 330, a fourth lens element 340, a fifth lens element 350, a sixth lens element 360, an IR-cut filter 370 and an image plane 380, wherein the imaging lens assembly has a total of six lens elements (310-360) with refractive power.

The first lens element 310 with positive refractive power has an object-side surface 311 being convex in a paraxial region thereof and an image-side surface 312 being concave in a paraxial region thereof. The first lens element 310 is made of plastic material and has the object-side surface 311 and the image-side surface 312 being both aspheric.

The second lens element 320 with positive refractive power has an object-side surface 321 being convex in a paraxial region thereof and an image-side surface 322 being concave in a paraxial region thereof, wherein the image-side surface 322 of the second lens element 320 has at least one convex shape in an off-axis region thereof. The second lens element 320 is made of plastic material and has the object-side surface 321 and the image-side surface 322 being both aspheric.

The third lens element 330 with positive refractive power has an object-side surface 331 being convex in a paraxial region thereof and an image-side surface 332 being convex in a paraxial region thereof. The third lens element 330 is made of plastic material and has the object-side surface 331 and the image-side surface 332 being both aspheric.

The fourth lens element 340 with negative refractive power has an object-side surface 341 being concave in a paraxial region thereof and an image-side surface 342 being convex in a paraxial region thereof. The fourth lens element 340 is made of plastic material and has the object-side surface 341 and the image-side surface 342 being both aspheric.

The fifth lens element 350 with positive refractive power has an object-side surface 351 being convex in a paraxial region thereof and an image-side surface 352 being concave in a paraxial region thereof. The fifth lens element 350 is made of plastic material and has the object-side surface 351 and the image-side surface 352 being both aspheric.

The sixth lens element 360 with negative refractive power has an object-side surface 361 being convex in a paraxial region thereof and an image-side surface 362 being concave in a paraxial region thereof, wherein the image-side surface 362 of the sixth lens element 360 has at least one convex shape in an off-axis region thereof. The sixth lens element 360 is made of plastic material and has the object-side surface 361 and the image-side surface 362 being both aspheric.

The IR-cut filter 370 is made of glass and located between the sixth lens element 360 and the image plane 380, and will not affect the focal length of the imaging lens assembly. The image sensor 390 is disposed on the image plane 380 of the imaging lens assembly.

The detailed optical data of the 3rd embodiment are shown in Table 5 and the aspheric surface data are shown in Table 6 below.

TABLE 5

3rd Embodiment							
f = 4.21 mm, Fno = 2.24, HFOV = 38.7 deg.							
Surface #		Curvature Radius	Thickness	Material	Index	Abbe #	Focal Length
0	Object	Plano	Infinity				
1	Ape. Stop	Plano	-0.321				
2	Lens 1	1.521 ASP	0.361	Plastic	1.544	55.9	12.17

TABLE 5-continued

3rd Embodiment f = 4.21 mm, Fno = 2.24, HFOV = 38.7 deg.								
Surface #		Curvature Radius	Thickness	Material	Index	Abbe #	Focal Length	
3		1.810 ASP	0.226					
4	Lens 2	1.817 ASP	0.355	Plastic	1.544	55.9	6.19	
5		3.674 ASP	0.155					
6	Lens 3	457.919 ASP	0.352	Plastic	1.544	55.9	4.89	
7		-2.673 ASP	0.123					
8	Lens 4	-1.347 ASP	0.844	Plastic	1.640	23.3	-3.67	
9		-3.939 ASP	0.117					
10	Lens 5	2.001 ASP	0.500	Plastic	1.544	55.9	5.39	
11		5.733 ASP	0.834					
12	Lens 6	13.982 ASP	0.380	Plastic	1.544	55.9	-4.43	
13		2.035 ASP	0.500					
14	IR-cut filter	Plano	0.210	Glass	1.517	64.2	—	
15		Plano	0.295					
16	Image	Plano	—					

Note:

Reference wavelength is 587.6 nm (d-line).

TABLE 6

Aspheric Coefficients						
Surface #	2	3	4	5	6	7
k =	5.4726E-01	1.3077E+00	-6.1311E+00	-1.7438E+01	-2.1551E+00	-8.2908E+00
A4 =	-2.4761E-02	-1.1839E-01	-3.0212E-02	-8.5933E-02	-1.3194E-01	-8.0639E-02
A6 =	-7.3143E-02	-2.0234E-03	8.6577E-03	-1.8836E-01	-4.4740E-02	-1.1050E-01
A8 =	3.3950E-01	1.1113E-01	-5.8136E-01	4.5294E-02	-2.8077E-01	1.6801E-01
A10 =	-7.1651E-01	-3.2925E-01	1.3202E+00	-1.4841E-01	2.3831E-01	-2.6626E-02
A12 =	7.2937E-01	3.3859E-01	-1.6418E+00	2.2212E-01	3.5278E-01	-6.2682E-02
A14 =	-3.0317E-01	-2.0969E-01	7.2561E-01	-4.8987E-03	-2.7910E-01	3.6509E-02
Surface #	8	9	10	11	12	13
k =	-9.4518E-01	-4.8701E-01	-1.4579E+01	-2.8676E+00	2.4836E+01	-9.7908E+00
A4 =	1.1101E-01	-8.8554E-02	4.1595E-02	4.6531E-02	-1.7891E-01	-9.2937E-02
A6 =	-2.1823E-01	1.7050E-01	-6.0384E-02	-6.0074E-02	1.0689E-01	4.6012E-02
A8 =	5.8895E-01	-1.3320E-01	3.9784E-02	3.3119E-02	-4.8477E-02	-1.7379E-02
A10 =	-8.0051E-01	7.5254E-02	-1.5741E-02	-1.0411E-02	1.4423E-02	4.2118E-03
A12 =	5.4346E-01	-2.3208E-02	3.6146E-03	1.8754E-03	-2.4341E-03	-6.2793E-04
A14 =	-1.4081E-01	2.7275E-03	-4.4886E-04	-1.7890E-04	2.1147E-04	5.1901E-05
A16 =			2.2916E-05	6.8951E-06	-7.3929E-06	-1.7890E-06

In the 3rd embodiment, the equation of the aspheric surface profiles of the aforementioned lens elements is the same as the equation of the 1st embodiment. Also, the definitions of these parameters shown in the following table are the same as those stated in the 1st embodiment with corresponding values for the 3rd embodiment, so an explanation in this regard will not be provided again.

Moreover, these parameters can be calculated from Table 5 and Table 6 as the following values and satisfy the following conditions:

3rd Embodiment			
f [mm]	4.21	R2 /R1	1.19
Fno	2.24	R12/f	0.48
HFOV [deg.]	38.7	f/f3	0.86
CT4/CT3	2.40	f2/f3	1.27
T56/(T12 + T23 + T34 + T45)	1.34	f3/f1 + f3/f2	1.19
Sag42 /CT4	0.11		

4th Embodiment

FIG. 4A is a schematic view of an imaging device according to the 4th embodiment of the present disclosure. FIG. 4B shows, in order from left to right, spherical aberration curves, astigmatic field curves and a distortion curve of the imaging device according to the 4th embodiment. In FIG. 4A, the imaging device includes the imaging lens assembly (not otherwise herein labeled) of the present disclosure and an image sensor 490. The imaging lens assembly includes, in order from an object side to an image side, an aperture stop 400, a first lens element 410, a second lens element 420, a third lens element 430, a fourth lens element 440, a fifth lens element 450, a sixth lens element 460, an IR-cut filter 470 and an image plane 480, wherein the imaging lens assembly has a total of six lens elements (410-460) with refractive power.

The first lens element 410 with positive refractive power has an object-side surface 411 being convex in a paraxial region thereof and an image-side surface 412 being concave in a paraxial region thereof. The first lens element 410 is made of plastic material and has the object-side surface 411 and the image-side surface 412 being both aspheric.

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The second lens element **420** with positive refractive power has an object-side surface **421** being convex in a paraxial region thereof and an image-side surface **422** being concave in a paraxial region thereof, wherein the image-side surface **422** of the second lens element **420** has at least one convex shape in an off-axis region thereof. The second lens element **420** is made of plastic material and has the object-side surface **421** and the image-side surface **422** being both aspheric.

The third lens element **430** with positive refractive power has an object-side surface **431** being convex in a paraxial region thereof and an image-side surface **432** being convex in a paraxial region thereof. The third lens element **430** is made of plastic material and has the object-side surface **431** and the image-side surface **432** being both aspheric.

The fourth lens element **440** with negative refractive power has an object-side surface **441** being concave in a paraxial region thereof and an image-side surface **442** being convex in a paraxial region thereof. The fourth lens element **440** is made of plastic material and has the object-side surface **441** and the image-side surface **442** being both aspheric.

The fifth lens element **450** with positive refractive power has an object-side surface **451** being convex in a paraxial

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region thereof and an image-side surface **452** being concave in a paraxial region thereof. The fifth lens element **450** is made of plastic material and has the object-side surface **451** and the image-side surface **452** being both aspheric.

The sixth lens element **460** with negative refractive power has an object-side surface **461** being convex in a paraxial region thereof and an image-side surface **462** being concave in a paraxial region thereof, wherein the image-side surface **462** of the sixth lens element **460** has at least one convex shape in an off-axis region thereof. The sixth lens element **460** is made of plastic material and has the object-side surface **461** and the image-side surface **462** being both aspheric.

The IR-cut filter **470** is made of glass and located between the sixth lens element **460** and the image plane **480**, and will not affect the focal length of the imaging lens assembly. The image sensor **490** is disposed on the image plane **480** of the imaging lens assembly.

The detailed optical data of the 4th embodiment are shown in Table 7 and the aspheric surface data are shown in Table 8 below.

TABLE 7

4th Embodiment							
f = 4.25 mm, Fno = 2.24, HFOV = 38.7 deg.							
Surface #		Curvature Radius	Thickness	Material	Index	Abbe #	Focal Length
0	Object	Plano	Infinity				
1	Ape. Stop	Plano	-0.313				
2	Lens 1	1.553 ASP	0.369	Plastic	1.544	55.9	12.03
3		1.867 ASP	0.214				
4	Lens 2	1.848 ASP	0.356	Plastic	1.544	55.9	6.34
5		3.711 ASP	0.142				
6	Lens 3	48.685 ASP	0.333	Plastic	1.544	55.9	5.00
7		-2.878 ASP	0.120				
8	Lens 4	-1.416 ASP	0.854	Plastic	1.640	23.3	-3.82
9		-4.154 ASP	0.118				
10	Lens 5	2.014 ASP	0.500	Plastic	1.544	55.9	5.63
11		5.365 ASP	0.819				
12	Lens 6	14.248 ASP	0.415	Plastic	1.544	55.9	-4.29
13		1.987 ASP	0.500				
14	IR-cut filter	Plano	0.210	Glass	1.517	64.2	—
15		Plano	0.298				
16	Image	Plano	—				

Note:

Reference wavelength is 587.6 nm (d-line).

TABLE 8

Aspheric Coefficients						
Surface #	2	3	4	5	6	7
k =	6.1852E-01	1.1613E+00	-6.6738E+00	-2.2004E+01	-2.1551E+00	-8.9132E+00
A4 =	-3.0507E-02	-1.3002E-01	-4.1031E-02	-7.6725E-02	-9.9449E-02	-5.3530E-02
A6 =	-5.6718E-02	2.8639E-02	-8.1545E-02	-3.1549E-01	-1.2999E-01	-1.1843E-01
A8 =	2.7660E-01	3.7701E-02	-1.7345E-01	3.4136E-01	-2.5851E-01	-1.1594E-02
A10 =	-5.9173E-01	-2.0435E-01	3.5222E-01	-5.3762E-01	4.6967E-01	3.3149E-01
A12 =	5.9845E-01	2.3804E-01	-4.6040E-01	6.2658E-01	-4.8799E-02	-3.5174E-01
A14 =	-2.4773E-01	-1.5941E-01	2.0772E-01	-2.1043E-01	-8.8108E-02	1.2808E-01
Surface #	8	9	10	11	12	13
k =	-9.1185E-01	-7.0392E-01	-1.4292E+01	-4.7274E+00	2.5706E+01	-9.6203E+00
A4 =	1.0170E-01	-1.0207E-01	1.9350E-02	3.3697E-02	-1.9429E-01	-1.0101E-01
A6 =	-1.8128E-01	1.8863E-01	-3.3744E-02	-4.4985E-02	1.2202E-01	5.4207E-02
A8 =	3.7103E-01	-1.5581E-01	1.9360E-02	2.4675E-02	-5.4497E-02	-2.0944E-02
A10 =	-3.6408E-01	1.0079E-01	-5.5211E-03	-7.5916E-03	1.5800E-02	5.1372E-03
A12 =	2.0846E-01	-3.6192E-02	4.6265E-04	1.2811E-03	-2.6461E-03	-7.8056E-04
A14 =	-5.2941E-02	4.9862E-03	8.2920E-05	-1.0694E-04	2.3271E-04	6.6228E-05
A16 =			-1.4345E-05	3.1672E-06	-8.3812E-06	-2.3514E-06

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In the 4th embodiment, the equation of the aspheric surface profiles of the aforementioned lens elements is the same as the equation of the 1st embodiment. Also, the definitions of these parameters shown in the following table are the same as those stated in the 1st embodiment with corresponding values for the 4th embodiment, so an explanation in this regard will not be provided again.

Moreover, these parameters can be calculated from Table 7 and Table 8 as the following values and satisfy the following conditions:

4th Embodiment			
f [mm]	4.25	R2 /R1	1.20
Fno	2.24	R12/f	0.47
HFOV [deg.]	38.7	f/f3	0.85
CT4/CT3	2.56	f2/f3	1.27
T56/(T12 + T23 + T34 + T45)	1.38	f3/f1 + f3/f2	1.20
Sag42 /CT4	0.09		

5th Embodiment

FIG. 5A is a schematic view of an imaging device according to the 5th embodiment of the present disclosure. FIG. 5B shows, in order from left to right, spherical aberration curves, astigmatic field curves and a distortion curve of the imaging device according to the 5th embodiment. In FIG. 5A, the imaging device includes the imaging lens assembly (not otherwise herein labeled) of the present disclosure and an image sensor 590. The imaging lens assembly includes, in order from an object side to an image side, an aperture stop 500, a first lens element 510, a second lens element 520, a third lens element 530, a fourth lens element 540, a fifth lens element 550, a sixth lens element 560, an IR-cut filter 570 and an image plane 580, wherein the imaging lens assembly has a total of six lens elements (510-560) with refractive power.

The first lens element 510 with positive refractive power has an object-side surface 511 being convex in a paraxial region thereof and an image-side surface 512 being concave in a paraxial region thereof. The first lens element 510 is made of plastic material and has the object-side surface 511 and the image-side surface 512 being both aspheric.

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The second lens element 520 with positive refractive power has an object-side surface 521 being convex in a paraxial region thereof and an image-side surface 522 being concave in a paraxial region thereof, wherein the image-side surface 522 of the second lens element 520 has at least one convex shape in an off-axis region thereof. The second lens element 520 is made of plastic material and has the object-side surface 521 and the image-side surface 522 being both aspheric.

The third lens element 530 with positive refractive power has an object-side surface 531 being convex in a paraxial region thereof and an image-side surface 532 being convex in a paraxial region thereof. The third lens element 530 is made of plastic material and has the object-side surface 531 and the image-side surface 532 being both aspheric.

The fourth lens element 540 with negative refractive power has an object-side surface 541 being concave in a paraxial region thereof and an image-side surface 542 being convex in a paraxial region thereof. The fourth lens element 540 is made of plastic material and has the object-side surface 541 and the image-side surface 542 being both aspheric.

The fifth lens element 550 with positive refractive power has an object-side surface 551 being convex in a paraxial region thereof and an image-side surface 552 being concave in a paraxial region thereof. The fifth lens element 550 is made of plastic material and has the object-side surface 551 and the image-side surface 552 being both aspheric.

The sixth lens element 560 with negative refractive power has an object-side surface 561 being convex in a paraxial region thereof and an image-side surface 562 being concave in a paraxial region thereof, wherein the image-side surface 562 of the sixth lens element 560 has at least one convex shape in an off-axis region thereof. The sixth lens element 560 is made of plastic material and has the object-side surface 561 and the image-side surface 562 being both aspheric.

The IR-cut filter 570 is made of glass and located between the sixth lens element 560 and the image plane 580, and will not affect the focal length of the imaging lens assembly. The image sensor 590 is disposed on the image plane 580 of the imaging lens assembly.

The detailed optical data of the 5th embodiment are shown in Table 9 and the aspheric surface data are shown in Table 10 below.

TABLE 9

5th Embodiment							
f = 4.03 mm, Fno = 2.00, HFOV = 39.5 deg.							
Surface #		Curvature Radius	Thickness	Material	Index	Abbe #	Focal Length
0	Object	Plano	Infinity				
1	Ape. Stop	Plano	-0.349				
2	Lens 1	1.621 ASP	0.414	Plastic	1.544	55.9	11.52
3		1.990 ASP	0.228				
4	Lens 2	1.946 ASP	0.372	Plastic	1.544	55.9	7.92
5		3.309 ASP	0.102				
6	Lens 3	12.664 ASP	0.377	Plastic	1.544	55.9	4.22
7		-2.774 ASP	0.158				
8	Lens 4	-1.353 ASP	0.763	Plastic	1.632	23.4	-3.67
9		-3.957 ASP	0.101				
10	Lens 5	2.152 ASP	0.578	Plastic	1.535	55.7	5.46
11		7.396 ASP	0.656				
12	Lens 6	6.698 ASP	0.530	Plastic	1.535	55.7	-4.45
13		1.708 ASP	0.400				
14	IR-cut filter	Plano	0.200	Glass	1.517	64.2	—
15		Plano	0.313				
16	Image	Plano	—				

Note:
Reference wavelength is 587.6 nm (d-line).

TABLE 10

Aspheric Coefficients						
Surface #	2	3	4	5	6	7
k =	9.1460E-01	1.5819E+00	-7.8084E+00	-1.7591E+01	-5.0000E+01	-3.5493E+00
A4 =	-3.8249E-02	-1.1023E-01	-2.6829E-02	-7.8868E-02	-9.6449E-02	-5.4217E-02
A6 =	-5.6473E-02	4.0095E-02	-5.6599E-02	-3.2530E-01	-1.3100E-01	-1.1104E-01
A8 =	3.0402E-01	3.3693E-02	-1.7077E-01	3.5502E-01	-2.7277E-01	-1.3593E-02
A10 =	-6.6615E-01	-2.2733E-01	3.6972E-01	-6.0426E-01	5.0243E-01	3.5445E-01
A12 =	6.3424E-01	2.6822E-01	-5.2400E-01	6.8518E-01	-6.9588E-02	-3.8754E-01
A14 =	-2.3915E-01	-1.5808E-01	2.1499E-01	-2.4923E-01	-9.2193E-02	1.3716E-01
Surface #	8	9	10	11	12	13
k =	-6.3233E-01	2.2743E+00	-1.5695E+01	-5.7995E+00	-1.4373E+01	-7.1181E+00
A4 =	8.8445E-02	-1.0749E-01	3.3079E-03	3.2350E-02	-2.0307E-01	-1.0076E-01
A6 =	-1.9183E-01	1.9455E-01	-3.0099E-02	-4.6618E-02	1.2727E-01	5.6468E-02
A8 =	4.0080E-01	-1.6576E-01	1.9885E-02	2.6195E-02	-5.7878E-02	-2.2205E-02
A10 =	-3.9576E-01	1.1017E-01	-6.4615E-03	-8.2183E-03	1.7098E-02	5.5451E-03
A12 =	2.2467E-01	-3.9367E-02	5.6136E-04	1.4052E-03	-2.9112E-03	-8.6033E-04
A14 =	-5.2511E-02	5.3425E-03	1.5143E-04	-1.1899E-04	2.6065E-04	7.4070E-05
A16 =			-3.3605E-05	3.6641E-06	-9.6125E-06	-2.6572E-06

In the 5th embodiment, the equation of the aspheric surface profiles of the aforementioned lens elements is the same as the equation of the 1st embodiment. Also, the definitions of these parameters shown in the following table are the same as those stated in the 1st embodiment with corresponding values for the 5th embodiment, so an explanation in this regard will not be provided again.

Moreover, these parameters can be calculated from Table 9 and Table 10 as the following values and satisfy the following conditions:

5th Embodiment			
f [mm]	4.03	R2 /R1	1.23
Fno	2.00	R12/f	0.42
HFOV [deg.]	39.5	f/f3	0.95
CT4/CT3	2.02	f2/f3	1.88
T56/(T12 + T23 + T34 + T45)	1.11	f3/f1 + f3/f2	0.90
Sag42 /CT4	0.15		

6th Embodiment

FIG. 6A is a schematic view of an imaging device according to the 6th embodiment of the present disclosure. FIG. 6B shows, in order from left to right, spherical aberration curves, astigmatic field curves and a distortion curve of the imaging device according to the 6th embodiment. In FIG. 6A, the imaging device includes the imaging lens assembly (not otherwise herein labeled) of the present disclosure and an image sensor 690. The imaging lens assembly includes, in order from an object side to an image side, an aperture stop 600, a first lens element 610, a second lens element 620, a third lens element 630, a fourth lens element 640, a fifth lens element 650, a sixth lens element 660, an IR-cut filter 670 and an image plane 680, wherein the imaging lens assembly has a total of six lens elements (610-660) with refractive power.

The first lens element 610 with positive refractive power has an object-side surface 611 being convex in a paraxial region thereof and an image-side surface 612 being concave in a paraxial region thereof. The first lens element 610 is made of plastic material and has the object-side surface 611 and the image-side surface 612 being both aspheric.

The second lens element 620 with positive refractive power has an object-side surface 621 being convex in a paraxial region thereof and an image-side surface 622 being concave in a paraxial region thereof, wherein the image-side surface 622 of the second lens element 620 has at least one convex shape in an off-axis region thereof. The second lens element 620 is made of plastic material and has the object-side surface 621 and the image-side surface 622 being both aspheric.

The third lens element 630 with positive refractive power has an object-side surface 631 being convex in a paraxial region thereof and an image-side surface 632 being convex in a paraxial region thereof. The third lens element 630 is made of plastic material and has the object-side surface 631 and the image-side surface 632 being both aspheric.

The fourth lens element 640 with negative refractive power has an object-side surface 641 being concave in a paraxial region thereof and an image-side surface 642 being convex in a paraxial region thereof. The fourth lens element 640 is made of plastic material and has the object-side surface 641 and the image-side surface 642 being both aspheric.

The fifth lens element 650 with positive refractive power has an object-side surface 651 being convex in a paraxial region thereof and an image-side surface 652 being concave in a paraxial region thereof. The fifth lens element 650 is made of plastic material and has the object-side surface 651 and the image-side surface 652 being both aspheric.

The sixth lens element 660 with negative refractive power has an object-side surface 661 being convex in a paraxial region thereof and an image-side surface 662 being concave in a paraxial region thereof, wherein the image-side surface 662 of the sixth lens element 660 has at least one convex shape in an off-axis region thereof. The sixth lens element 660 is made of plastic material and has the object-side surface 661 and the image-side surface 662 being both aspheric.

The IR-cut filter 670 is made of glass and located between the sixth lens element 660 and the image plane 680, and will not affect the focal length of the imaging lens assembly. The image sensor 690 is disposed on the image plane 680 of the imaging lens assembly.

The detailed optical data of the 6th embodiment are shown in Table 11 and the aspheric surface data are shown in Table 12 below.

TABLE 11

6th Embodiment f = 3.35 mm, Fno = 1.95, HFOV = 41.0 deg.							
Surface #		Curvature Radius	Thickness	Material	Index	Abbe #	Focal Length
0	Object	Plano	Infinity				
1	Ape. Stop	Plano	-0.260				
2	Lens 1	1.522 ASP	0.351	Plastic	1.544	55.9	11.62
3		1.841 ASP	0.158				
4		1.861 ASP	0.380				
5	Lens 2	3.674 ASP	0.082	Plastic	1.544	55.9	6.45
6		16.606 ASP	0.368				
7		-2.256 ASP	0.153				
8	Lens 3	-1.101 ASP	0.559	Plastic	1.640	23.3	-3.19
9		-2.866 ASP	0.093				
10		1.946 ASP	0.647				
11	Lens 4	8.214 ASP	0.583	Plastic	1.535	55.7	-3.87
12		3.936 ASP	0.406				
13		1.308 ASP	0.400				
14	IR-cut filter	Plano	0.200	Glass	1.517	64.2	—
15		Plano	0.125				
16	Image	Plano	—				

Note:

Reference wavelength is 587.6 nm (d-line).

TABLE 12

Aspheric Coefficients						
Surface #	2	3	4	5	6	7
k =	1.0636E+00	1.3764E+00	-8.4810E+00	-1.8723E+01	-2.7062E+01	-1.7521E+00
A4 =	-5.3137E-02	-1.6373E-01	-4.4736E-02	-1.1712E-01	-1.4025E-01	-8.8506E-02
A6 =	-1.1947E-01	7.6924E-02	-9.6572E-02	-6.1151E-01	-2.3938E-01	-2.0951E-01
A8 =	7.4113E-01	6.4667E-02	-3.9059E-01	8.5487E-01	-6.6743E-01	-5.0684E-02
A10 =	-2.0443E+00	-7.0993E-01	1.1305E+00	-1.8442E+00	1.5183E+00	1.0855E+00
A12 =	2.4993E+00	1.0760E+00	-2.1312E+00	2.7184E+00	-3.0014E-01	-1.5281E+00
A14 =	-1.2164E+00	-8.1307E-01	1.1122E+00	-1.3353E+00	-4.2626E-01	7.1657E-01
Surface #	8	9	10	11	12	13
k =	-5.8361E-01	1.9753E+00	-1.8087E+01	-2.8489E+00	-1.2530E+01	-5.9753E+00
A4 =	1.2130E-01	-1.5757E-01	-5.8618E-03	4.3260E-02	-3.2015E-01	-1.5457E-01
A6 =	-3.6570E-01	3.6917E-01	-5.3021E-02	-8.9283E-02	2.3995E-01	1.0814E-01
A8 =	9.6449E-01	-3.9710E-01	4.6976E-02	6.3417E-02	-1.3859E-01	-5.3669E-02
A10 =	-1.2184E+00	3.4027E-01	-2.1002E-02	-2.5195E-02	5.2729E-02	1.7059E-02
A12 =	8.9773E-01	-1.5538E-01	2.3981E-03	5.5337E-03	-1.1548E-02	-3.4008E-03
A14 =	-2.5801E-01	2.7270E-02	1.0468E-03	-6.1475E-04	1.3241E-03	3.7754E-04
A16 =			-3.6211E-04	2.3855E-05	-6.2599E-05	-1.7422E-05

In the 6th embodiment, the equation of the aspheric surface profiles of the aforementioned lens elements is the same as the equation of the 1st embodiment. Also, the definitions of these parameters shown in the following table are the same as those stated in the 1st embodiment with corresponding values for the 6th embodiment, so an explanation in this regard will not be provided again.

Moreover, these parameters can be calculated from Table 11 and Table 12 as the following values and satisfy the following conditions:

6th Embodiment			
f [mm]	3.35	R2 /R1	1.21
Fno	1.95	R12/f	0.39
HFOV [deg.]	41.0	f/f3	0.91
CT4/CT3	1.52	f2/f3	1.75
T56/(T12 + T23 + T34 + T45)	1.20	f3/f1 + f3/f2	0.89
Sag42 /CT4	0.26		

7th Embodiment

FIG. 7A is a schematic view of an imaging device according to the 7th embodiment of the present disclosure. FIG. 7B shows, in order from left to right, spherical aberration curves, astigmatic field curves and a distortion curve of the imaging device according to the 7th embodiment. In FIG. 7A, the imaging device includes the imaging lens assembly (not otherwise herein labeled) of the present disclosure and an image sensor 790. The imaging lens assembly includes, in order from an object side to an image side, a first lens element 710, an aperture stop 700, a second lens element 720, a third lens element 730, a fourth lens element 740, a fifth lens element 750, a sixth lens element 760, an IR-cut filter 770 and an image plane 780, wherein the imaging lens assembly has a total of six lens elements (710-760) with refractive power.

The first lens element 710 with negative refractive power has an object-side surface 711 being convex in a paraxial region thereof and an image-side surface 712 being concave in a paraxial region thereof. The first lens element 710 is made of glass material and has the object-side surface 711 and the image-side surface 712 being both aspheric.

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The second lens element **720** with positive refractive power has an object-side surface **721** being convex in a paraxial region thereof and an image-side surface **722** being concave in a paraxial region thereof, wherein the image-side surface **722** of the second lens element **720** has at least one convex shape in an off-axis region thereof. The second lens element **720** is made of plastic material and has the object-side surface **721** and the image-side surface **722** being both aspheric.

The third lens element **730** with positive refractive power has an object-side surface **731** being planar in a paraxial region thereof and an image-side surface **732** being convex in a paraxial region thereof. The third lens element **730** is made of plastic material and has the object-side surface **731** and the image-side surface **732** being both aspheric.

The fourth lens element **740** with negative refractive power has an object-side surface **741** being concave in a paraxial region thereof and an image-side surface **742** being convex in a paraxial region thereof. The fourth lens element **740** is made of plastic material and has the object-side surface **741** and the image-side surface **742** being both aspheric.

The fifth lens element **750** with positive refractive power has an object-side surface **751** being convex in a paraxial

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region thereof and an image-side surface **752** being concave in a paraxial region thereof. The fifth lens element **750** is made of plastic material and has the object-side surface **751** and the image-side surface **752** being both aspheric.

The sixth lens element **760** with negative refractive power has an object-side surface **761** being convex in a paraxial region thereof and an image-side surface **762** being concave in a paraxial region thereof, wherein the image-side surface **762** of the sixth lens element **760** has at least one convex shape in an off-axis region thereof. The sixth lens element **760** is made of plastic material and has the object-side surface **761** and the image-side surface **762** being both aspheric.

The IR-cut filter **770** is made of glass and located between the sixth lens element **760** and the image plane **780**, and will not affect the focal length of the imaging lens assembly. The image sensor **790** is disposed on the image plane **780** of the imaging lens assembly.

The detailed optical data of the 7th embodiment are shown in Table 13 and the aspheric surface data are shown in Table 14 below.

TABLE 13

7th Embodiment							
f = 4.04 mm, Fno = 2.35, HFOV = 39.7 deg.							
Surface #		Curvature Radius	Thickness	Material	Index	Abbe #	Focal Length
0	Object	Plano	Infinity				
1	Lens 1	1.638 ASP	0.352	Glass	1.542	62.9	-133.13
2		1.481 ASP	0.200				
3	Ape. Stop	Plano	-0.066				
4	Lens 2	1.480 ASP	0.410	Plastic	1.544	55.9	4.04
5		4.094 ASP	0.172				
6	Lens 3	∞ ASP	0.400	Plastic	1.544	55.9	4.91
7		-2.673 ASP	0.126				
8	Lens 4	-1.269 ASP	0.740	Plastic	1.639	23.5	-3.69
9		-3.374 ASP	0.172				
10	Lens 5	2.047 ASP	0.555	Plastic	1.544	55.9	4.56
11		10.604 ASP	0.775				
12	Lens 6	14.796 ASP	0.386	Plastic	1.535	55.7	-4.25
13		1.953 ASP	0.500				
14	IR-cut filter	Plano	0.210	Glass	1.517	64.2	—
15		Plano	0.312				
16	Image	Plano	—				

Note:

Reference wavelength is 587.6 nm (d-line).

TABLE 14

Aspheric Coefficients						
Surface #	1	2	4	5	6	7
k =	9.3204E-01	-6.2675E-01	-5.6314E+00	-1.8411E+01	-1.0000E+00	-8.2366E+00
A4 =	-5.9141E-02	-2.0688E-01	-2.9545E-02	-8.8579E-02	-1.4247E-01	-8.9235E-02
A6 =	-3.3653E-02	1.4799E-01	2.8388E-03	-1.9014E-01	-6.3930E-02	-1.2475E-01
A8 =	2.9295E-01	3.3886E-02	-5.3913E-01	6.5909E-02	-2.9749E-01	1.5778E-01
A10 =	-6.9881E-01	-3.2429E-01	1.2840E+00	-1.4636E-01	2.7538E-01	-2.5382E-02
A12 =	7.2970E-01	3.3761E-01	-1.6469E+00	2.2370E-01	3.5397E-01	-5.3508E-02
A14 =	-3.0317E-01	-2.0969E-01	7.2561E-01	-4.9512E-03	-2.7736E-01	3.6011E-02
Surface #	8	9	10	11	12	13
k =	-1.0487E+00	-2.1803E+00	-1.4448E+01	-2.9181E+00	2.0834E+01	-9.3688E+00
A4 =	1.1575E-01	-8.4700E-02	2.1864E-02	2.8569E-02	-1.7755E-01	-8.4873E-02
A6 =	-2.1046E-01	1.7099E-01	-1.6164E-02	-1.3360E-02	1.1422E-01	4.2568E-02
A8 =	5.9242E-01	-1.3293E-01	-9.3716E-04	-7.0042E-03	-5.8412E-02	-1.7670E-02
A10 =	-8.0580E-01	7.5439E-02	3.4849E-03	6.7956E-03	1.9302E-02	4.6269E-03
A12 =	5.3700E-01	-2.3177E-02	-1.1943E-03	-2.0434E-03	-3.5779E-03	-7.1008E-04
A14 =	-1.3524E-01	2.6689E-03	1.4738E-04	2.7438E-04	3.4131E-04	5.8214E-05
A16 =			-5.2919E-06	-1.4069E-05	-1.3129E-05	-1.9588E-06

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In the 7th embodiment, the equation of the aspheric surface profiles of the aforementioned lens elements is the same as the equation of the 1st embodiment. Also, the definitions of these parameters shown in the following table are the same as those stated in the 1st embodiment with corresponding values for the 7th embodiment, so an explanation in this regard will not be provided again.

Moreover, these parameters can be calculated from Table 13 and Table 14 as the following values and satisfy the following conditions:

7th Embodiment			
f [mm]	4.04	R2 /R1	0.90
Fno	2.35	R12/f	0.48
HFOV [deg.]	39.7	f/f3	0.82
CT4/CT3	1.85	f2/f3	0.82
T56/(T12 + T23 + T34 + T45)	1.28	f3/f1 + f3/f2	1.25
Sag42 /CT4	0.15		

8th Embodiment

FIG. 8A is a schematic view of an imaging device according to the 8th embodiment of the present disclosure. FIG. 8B shows, in order from left to right, spherical aberration curves, astigmatic field curves and a distortion curve of the imaging device according to the 8th embodiment. In FIG. 8A, the imaging device includes the imaging lens assembly (not otherwise herein labeled) of the present disclosure and an image sensor 890. The imaging lens assembly includes, in order from an object side to an image side, an aperture stop 800, a first lens element 810, a second lens element 820, a third lens element 830, a fourth lens element 840, a fifth lens element 850, a sixth lens element 860, an IR-cut filter 870 and an image plane 880, wherein the imaging lens assembly has a total of six lens elements (810-860) with refractive power.

The first lens element 810 with positive refractive power has an object-side surface 811 being convex in a paraxial region thereof and an image-side surface 812 being concave in a paraxial region thereof. The first lens element 810 is made

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of plastic material and has the object-side surface 811 and the image-side surface 812 being both aspheric.

The second lens element 820 with positive refractive power has an object-side surface 821 being convex in a paraxial region thereof and an image-side surface 822 being convex in a paraxial region thereof. The second lens element 820 is made of plastic material and has the object-side surface 821 and the image-side surface 822 being both aspheric.

The third lens element 830 with positive refractive power has an object-side surface 831 being concave in a paraxial region thereof and an image-side surface 832 being convex in a paraxial region thereof. The third lens element 830 is made of plastic material and has the object-side surface 831 and the image-side surface 832 being both aspheric.

The fourth lens element 840 with negative refractive power has an object-side surface 841 being concave in a paraxial region thereof and an image-side surface 842 being convex in a paraxial region thereof. The fourth lens element 840 is made of plastic material and has the object-side surface 841 and the image-side surface 842 being both aspheric.

The fifth lens element 850 with positive refractive power has an object-side surface 851 being convex in a paraxial region thereof and an image-side surface 852 being convex in a paraxial region thereof. The fifth lens element 850 is made of plastic material and has the object-side surface 851 and the image-side surface 852 being both aspheric.

The sixth lens element 860 with negative refractive power has an object-side surface 861 being convex in a paraxial region thereof and an image-side surface 862 being concave in a paraxial region thereof, wherein the image-side surface 862 of the sixth lens element 860 has at least one convex shape in an off-axis region thereof. The sixth lens element 860 is made of plastic material and has the object-side surface 861 and the image-side surface 862 being both aspheric.

The IR-cut filter 870 is made of glass and located between the sixth lens element 860 and the image plane 880, and will not affect the focal length of the imaging lens assembly. The image sensor 890 is disposed on the image plane 880 of the imaging lens assembly.

The detailed optical data of the 8th embodiment are shown in Table 15, and the aspheric surface data are shown in Table 16 below.

TABLE 15

8th Embodiment							
f = 3.55 mm, Fno = 2.45, HFOV = 39.7 deg.							
Surface #		Curvature Radius	Thickness	Material	Index	Abbe #	Focal Length
0	Object	Plano	Infinity				
1	Ape. Stop	Plano	-0.158				
2	Lens 1	1.545 ASP	0.228	Plastic	1.544	55.9	45.26
3		1.563 ASP	0.142				
4	Lens 2	1.768 ASP	0.511	Plastic	1.544	55.9	3.20
5		-100.000 ASP	0.188				
6	Lens 3	-4.514 ASP	0.342	Plastic	1.544	55.9	4.95
7		-1.732 ASP	0.083				
8	Lens 4	-1.001 ASP	0.478	Plastic	1.639	23.5	-2.77
9		-2.737 ASP	0.170				
10	Lens 5	2.424 ASP	0.506	Plastic	1.607	26.6	3.51
11		-16.291 ASP	0.573				
12	Lens 6	10.996 ASP	0.381	Plastic	1.583	30.2	-3.24
13		1.593 ASP	0.500				
14	IR-cut filter	Plano	0.175	Glass	1.517	64.2	—
15		Plano	0.329				
16	Image	Plano	—				

Note:

Reference wavelength is 587.6 nm (d-line).

TABLE 16

Aspheric Coefficients						
Surface #	2	3	4	5	6	7
k =	9.7974E-01	-5.6645E-01	-6.9743E+00	9.0000E+01	2.4933E+00	-3.2285E+00
A4 =	-1.0210E-01	-1.9123E-01	-2.5447E-02	-1.4095E-01	-1.8706E-01	-1.3029E-01
A6 =	-4.6920E-02	1.0025E-01	-1.5606E-03	-1.8539E-01	7.9731E-02	1.3110E-01
A8 =	3.0832E-01	1.5237E-02	-5.7169E-01	6.2672E-02	-7.1721E-01	-4.3165E-01
A10 =	-7.5590E-01	-2.7719E-01	1.2499E+00	-1.6014E-01	8.1414E-01	6.6072E-01
A12 =	7.4161E-01	3.2886E-01	-1.6522E+00	2.0120E-01	8.1855E-02	-4.4904E-01
A14 =	-3.0310E-01	-2.2103E-01	7.3139E-01	-3.3343E-02	-2.3330E-01	1.3461E-01
Surface #	8	9	10	11	12	13
k =	-1.4315E+00	-5.3727E+00	-2.9810E+01	6.8631E+01	2.6401E+01	-7.1445E+00
A4 =	1.3615E-01	-5.1692E-02	8.2345E-02	4.6788E-02	-3.0454E-01	-1.6403E-01
A6 =	-2.3628E-01	4.0459E-02	-1.8539E-01	2.1497E-02	1.7035E-01	8.7454E-02
A8 =	7.0165E-01	8.6499E-02	1.9357E-01	-1.0702E-01	-4.4281E-02	-2.9910E-02
A10 =	-9.0176E-01	-4.5104E-02	-1.4274E-01	1.0222E-01	-1.1869E-02	4.8354E-03
A12 =	5.4971E-01	-4.1140E-03	7.3143E-02	-4.5443E-02	1.2309E-02	2.7079E-05
A14 =	-1.2838E-01	4.4537E-03	-2.2542E-02	9.8321E-03	-3.1321E-03	-1.2253E-04
A16 =			2.8785E-03	-8.3982E-04	2.6606E-04	1.2138E-05

In the 8th embodiment, the equation of the aspheric surface profiles of the aforementioned lens elements is the same as the equation of the 1st embodiment. Also, the definitions of these parameters shown in the following table are the same as those stated in the 1st embodiment with corresponding values for the 8th embodiment, so an explanation in this regard will not be provided again. Moreover, these parameters can be calculated from Table 15 and Table 16 as the following values and satisfy the following conditions:

8th Embodiment			
f [mm]	3.55	R2 /R1	1.01
Fno	2.45	R12/f	0.45
HFOV [deg.]	39.7	f/f3	0.72
CT4/CT3	1.40	f2/f3	0.65
T56/(T12 + T23 + T34 + T45)	0.98	f3/f1 + f3/f2	1.66
Sag42 /CT4	0.21		

The foregoing imaging device **901** may be installed in but not limited to a mobile terminal, including a smart phone **910**, a tablet personal computer **920** or a wearable device **930**. Moreover, the imaging device **901** includes an imaging lens assembly (not otherwise herein labeled) and an image sensor (not otherwise herein labeled), wherein the image sensor is disposed on an image plane of the imaging lens assembly.

The foregoing description, for purpose of explanation, has been described with reference to specific embodiments. It is to be noted that TABLES 1-16 show different data of the different embodiments; however, the data of the different embodiments are obtained from experiments. The embodiments were chosen and described in order to best explain the principles of the disclosure and its practical applications, to thereby enable others skilled in the art to best utilize the disclosure and various embodiments with various modifications as are suited to the particular use contemplated. The embodiments depicted above and the appended drawings are exemplary and are not intended to be exhaustive or to limit the scope of the present disclosure to the precise forms disclosed. Many modifications and variations are possible in view of the above teachings.

What is claimed is:

1. An imaging lens assembly comprising, in order from an object side to an image side:

- a first lens element having refractive power;
- a second lens element having refractive power;
- a third lens element having positive refractive power;
- a fourth lens element having negative refractive power;
- a fifth lens element with refractive power having an image-side surface being concave in a paraxial region thereof, wherein an object-side surface and the image-side surface of the fifth lens element are aspheric; and
- a sixth lens element with refractive power having an image-side surface being concave in a paraxial region thereof, wherein the image-side surface of the sixth lens element has at least one convex shape in an off-axis region thereof, and an object-side surface and the image-side surface of the sixth lens element are aspheric;

wherein the imaging lens assembly has a total of six lens elements with refractive power, a central thickness of the third lens element is CT3, a central thickness of the fourth lens element is CT4, a focal length of the imaging lens assembly is f, and a focal length of the third lens element is f3, the following conditions are satisfied:

$$1.40 < CT4/CT3; \text{ and}$$

$$0.35 < f/f3.$$

2. The imaging lens assembly of claim 1, wherein a distance in parallel with an optical axis from an axial vertex on the image-side surface of the fourth lens element to a maximum effective radius position on the image-side surface of the fourth lens element is Sag42, the central thickness of the fourth lens element is CT4, and the following condition is satisfied:

$$|Sag42|/CT4 < 0.3.$$

3. The imaging lens assembly of claim 1, wherein the image-side surface of the second lens element has at least one convex shape in an off-axis region thereof.

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4. The imaging lens assembly of claim 1, wherein an Abbe number of a lens element is V, only one lens element among all of the lens elements of the imaging lens assembly satisfies the following condition:

$$V < 30.$$

5. An imaging device, comprising:
the imaging lens assembly of claim 1; and
an image sensor, wherein the image sensor is located on an image plane of the imaging lens assembly.

6. A mobile terminal, comprising:
the imaging device of claim 5.

7. The imaging lens assembly of claim 1, wherein the third lens element has an image-side surface being convex in a paraxial region thereof.

8. The imaging lens assembly of claim 7, wherein the second lens element has an object-side surface being convex in a paraxial region thereof.

9. The imaging lens assembly of claim 8, wherein the object-side surface of the fifth lens element is convex in a paraxial region thereof, the fifth lens element has positive refractive power, and the sixth lens element has negative refractive power.

10. The imaging lens assembly of claim 8, wherein the central thickness of the third lens element is CT3, the central thickness of the fourth lens element is CT4, and the following condition is satisfied:

$$1.75 < CT4/CT3 < 3.50.$$

11. The imaging lens assembly of claim 8, wherein the focal length of the imaging lens assembly is f, the focal length of the third lens element is f3, and the following condition is satisfied:

$$0.60 < f/f3 < 1.50.$$

12. The imaging lens assembly of claim 8, wherein an axial distance between the first lens element and the second lens element is T12, an axial distance between the second lens element and the third lens element is T23, an axial distance between the third lens element and the fourth lens element is T34, an axial distance between the fourth lens element and the fifth lens element is T45, an axial distance between the fifth lens element and the sixth lens element is T56, and the following condition is satisfied:

$$0.75 < T56/(T12+T23+T34+T45) < 2.0.$$

13. An imaging lens assembly comprising, in order from an object side to an image side:

- a first lens element having refractive power;
- a second lens element having positive refractive power;
- a third lens element having positive refractive power;
- a fourth lens element having negative refractive power;
- a fifth lens element with refractive power having an object-side surface and an image-side surface of the fifth lens element are aspheric; and
- a sixth lens element with refractive power having an image-side surface being concave in a paraxial region thereof, wherein the image-side surface of the sixth lens element has at least one convex shape in an off-axis region thereof, and an object-side surface and the image-side surface of the sixth lens element are aspheric;

wherein the imaging lens assembly has a total of six lens elements with refractive power, a central thickness of the third lens element is CT3, a central thickness of the fourth lens element is CT4, a focal length of the imaging

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lens assembly is f, a focal length of the third lens element is f3, and the following conditions are satisfied:

$$1.20 < CT4/CT3; \text{ and}$$

$$0.35 < f/f3.$$

14. The imaging lens assembly of claim 13, wherein a focal length of the second lens element is f2, the focal length of the third lens element is f3, and the following condition is satisfied:

$$0.75 < f2/f3 < 2.0.$$

15. The imaging lens assembly of claim 13, wherein an axial distance between the first lens element and the second lens element is T12, an axial distance between the second lens element and the third lens element is T23, an axial distance between the third lens element and the fourth lens element is T34, an axial distance between the fourth lens element and the fifth lens element is T45, an axial distance between the fifth lens element and the sixth lens element is T56, and the following condition is satisfied:

$$0.75 < T56/(T12+T23+T34+T45) < 2.0.$$

16. The imaging lens assembly of claim 13, wherein a distance in parallel with an optical axis from an axial vertex on the image-side surface of the fourth lens element to a maximum effective radius position on the image-side surface of the fourth lens element is Sag42, the central thickness of the fourth lens element is CT4, and the following condition is satisfied:

$$|Sag42|/CT4 < 0.3.$$

17. An imaging device, comprising:

the imaging lens assembly of claim 13; and
an image sensor, wherein the image sensor is located on an image plane of the imaging lens assembly.

18. A mobile terminal, comprising:
the imaging device of claim 16.

19. The imaging lens assembly of claim 13, wherein a curvature radius of the object-side surface of the first lens element is R1, a curvature radius of the image-side surface of the first lens element is R2, and the following condition is satisfied:

$$0.80 < |R2|/R1.$$

20. The imaging lens assembly of claim 19, wherein the second lens element has an object-side surface being convex in a paraxial region thereof.

21. The imaging lens assembly of claim 20, wherein the focal length of the imaging lens assembly is f, a curvature radius of the image-side surface of the sixth lens element is R12, and the following condition is satisfied:

$$0.20 < R12/f < 0.75.$$

22. The imaging lens assembly of claim 20, wherein the central thickness of the third lens element is CT3, the central thickness of the fourth lens element is CT4, and the following condition is satisfied:

$$1.50 < CT4/CT3 < 4.0.$$

23. An imaging lens assembly comprising, in order from an object side to an image side:

- a first lens element having refractive power;
- a second lens element having refractive power;
- a third lens element having positive refractive power;
- a fourth lens element having negative refractive power;

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a fifth lens element having refractive power, wherein an object-side surface and an image-side surface of the fifth lens element are aspheric; and

a sixth lens element with refractive power having an image-side surface being concave in a paraxial region thereof, wherein the image-side surface of the sixth lens element has at least one convex shape in an off-axis region thereof, and an object-side surface and the image-side surface of the sixth lens element are aspheric;

wherein the imaging lens assembly has a total of six lens elements with refractive power, a central thickness of the third lens element is CT3, a central thickness of the fourth lens element is CT4, a focal length of the imaging lens assembly is f, a focal length of the third lens element is f3, and the following conditions are satisfied:

$$1.75 < CT4/CT3 < 3.50; \text{ and}$$

$$0.50 < f/f3.$$

24. The imaging lens assembly of claim 23, wherein an axial distance between the first lens element and the second lens element is T12, an axial distance between the second lens element and the third lens element is T23, an axial distance between the third lens element and the fourth lens element is T34, an axial distance between the fourth lens element and the

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fifth lens element is T45, an axial distance between the fifth lens element and the sixth lens element is T56, and the following condition is satisfied:

$$0.75 < T56/(T12+T23+T34+T45) < 2.0.$$

25. The imaging lens assembly of claim 23, wherein a focal length of the first lens element is f1, a focal length of the second lens element is f2, the focal length of the third lens element is f3, and the following condition is satisfied:

$$|f3/f1| + |f3/f2| < 1.5.$$

26. The imaging lens assembly of claim 23, wherein the focal length of the imaging lens assembly is f, a curvature radius of the image-side surface of the sixth lens element is R12, and the following condition is satisfied:

$$0.30 < R12/f < 0.75.$$

27. The imaging lens assembly of claim 23, wherein the first lens element has an object-side surface being convex in a paraxial region thereof and an image-side surface being concave in a paraxial region thereof.

28. The imaging lens assembly of claim 23, wherein an Abbe number of a lens element is V, only one lens element among all of the lens elements of the imaging lens assembly satisfies the following condition:

$$V < 30.$$

* * * * *

UNITED STATES PATENT AND TRADEMARK OFFICE
CERTIFICATE OF CORRECTION

PATENT NO. : 9,195,030 B2
APPLICATION NO. : 14/223770
DATED : November 24, 2015
INVENTOR(S) : Lin-Yao Liao et al.

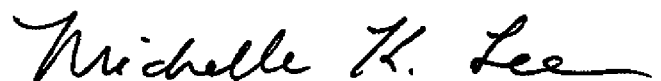
Page 1 of 1

It is certified that error appears in the above-identified patent and that said Letters Patent is hereby corrected as shown below:

In the Claims

Column 32, line 39 (claim 18): replace “the imaging device of claim 16.” with -- the imaging device of claim 17. --

Signed and Sealed this
Thirteenth Day of September, 2016

A handwritten signature in black ink, reading "Michelle K. Lee". The signature is fluid and cursive, with the first letters of each name being capitalized and prominent.

Michelle K. Lee
Director of the United States Patent and Trademark Office